Fundamental Modules - 5

WEEK 5: PROFESSIONAL PRACTICES & COURSE COMPLETION

Days 25-30 | Detailed Hour-by-Hour Structure

Week Theme: *"Working as a professional technician - quality, communication, and demonstrating competency"*

DAY 25: PREVENTIVE MAINTENANCE SCHEDULES + PM EXECUTION

Learning Objectives:

- Understand manufacturer service schedules
- Interpret service interval requirements
- Execute comprehensive preventive maintenance
- Perform multi-point inspection
- Document service properly

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Understanding Service Schedules

What is Preventive Maintenance?

- **Scheduled service before problems occur**
- Replace wear items on schedule
- Inspect for potential problems
- Adjust components as needed
- **Goal:** Prevent breakdowns, extend vehicle life, maintain safety

Why PM is Critical:

- **Reliability:** Reduces unexpected breakdowns
- **Safety:** Identifies safety issues early
- **Longevity:** Vehicles last longer with proper maintenance

- **Resale value:** Service records increase value
- **Fuel economy:** Well-maintained = efficient
- **Legal compliance:** Emission standards, safety inspections
- **Warranty protection:** Required for warranty validity
- **Cost savings:** Prevent expensive repairs

Manufacturer Service Schedules:

- Based on engineering testing and field data
- Specific to each vehicle model
- Found in owner's manual
- **Time-based OR mileage-based** (whichever comes first)

Typical Service Intervals:

- **Every 5,000 km / 3 months:** Oil change, basic inspection
- **Every 10,000 km / 6 months:** Minor service (oil, filters, inspection)
- **Every 20,000 km / 12 months:** Intermediate service (plus air filter, cabin filter)
- **Every 40,000 km / 24 months:** Major service (plus transmission fluid, coolant)
- **Every 60,000 km / 36 months:** Major service plus (spark plugs, belts)
- **Every 100,000 km:** Timing belt (if equipped), major overhaul items

Service Categories:

A-Service (Minor/Basic):

- Engine oil and filter change
- Visual inspection (fluids, belts, hoses, lights)
- Tire pressure check
- Brake inspection (visual)
- Reset service indicator
- **Time:** 45-60 minutes

B-Service (Major):

- All A-Service items
- Engine air filter replacement
- Cabin air filter replacement
- Brake fluid check
- Coolant level/condition check
- Comprehensive inspection all systems
- **Time:** 90-120 minutes

C-Service (Comprehensive):

- All B-Service items
- Transmission fluid change
- Coolant system service
- Spark plug replacement
- Drive belt replacement
- Brake fluid replacement
- Differential service (if applicable)
- **Time:** 2-4 hours

Hour 2 (10:00 AM - 11:00 AM): Service Schedule Interpretation

Reading Service Schedules:

Interval Format:

- "10,000 km OR 6 months, whichever comes first"
- Both time AND mileage matter
- Short trips, harsh conditions may require more frequent service

Service Classifications:

Normal Service:

- Typical daily driving
- Mix of city and highway
- Moderate climate
- Paved roads
- Standard intervals apply

Severe Service:

- **Requires MORE FREQUENT service**
- Conditions include:
  - **Short trips** (< 10 km per trip)
  - **Stop-and-go traffic** (extended idling)
  - **Extreme temperatures** (very hot or very cold)
  - **Dusty/dirty conditions**
  - **Towing or carrying heavy loads**
  - **Mountain driving** (steep grades)
  - **Off-road use**
  - **Commercial use** (taxi, delivery)

Severe Service Adjustments:

- Oil changes: Every 3,000-5,000 km (vs 10,000 km normal)
- Air filter: Every 10,000 km (vs 20,000 km)
- Transmission fluid: Every 20,000 km (vs 40,000 km)
- Brake inspections: More frequent
- **Many Indian driving conditions = severe service!**

Critical Timing-Based Services:

- **Timing belt:** 60,000-100,000 km
  - **Failure = engine damage** (interference engines)
  - Must not be delayed
  - Expensive but critical
- **Coolant:** 2-5 years
  - Corrosion inhibitors deplete
  - Not just about level
- **Brake fluid:** 2-3 years
  - Absorbs moisture over time
  - Boiling point decreases
- **Transmission fluid:** 40,000-80,000 km
  - Or "lifetime" (often means 150,000 km)
  - Critical for transmission longevity

Service Items by System:

Engine:

- Oil and filter: 5,000-10,000 km
- Air filter: 20,000-40,000 km
- Spark plugs: 40,000-100,000 km (depends on type)
- PCV valve: 60,000-100,000 km
- Fuel filter: 40,000-80,000 km (if replaceable)
- Timing belt: 60,000-100,000 km

Cooling System:

- Coolant: 40,000-100,000 km or 2-5 years
- Thermostat: As needed (if stuck)
- Hoses: Inspect annually, replace if cracked
- Water pump: 100,000-150,000 km (often with timing belt)

Brakes:

- Pads/shoes: 30,000-70,000 km (varies greatly)
- Rotors/drums: 80,000-150,000 km (if not damaged)
- Brake fluid: Every 2-3 years
- Calipers: As needed (if seized)

Chassis:

- Tires: 40,000-80,000 km (depends on quality, driving)
- Wheel alignment: Annually or if pulling
- Wheel balancing: With tire rotation or if vibration
- Suspension components: Inspect annually, replace as needed
- Steering components: Inspect annually, replace as needed

Electrical:

- Battery: 3-5 years
- Alternator: 100,000-150,000 km (if no issues)
- Starter: 100,000-200,000 km (if no issues)
- Belts: 60,000-100,000 km

Hour 3 (11:00 AM - 12:00 PM): Service Documentation Requirements

Why Documentation Matters:

- **Legal protection:** Proof service was performed correctly
- **Warranty compliance:** Required for warranty claims
- **Customer trust:** Professional appearance
- **Vehicle history:** Helps future technicians
- **Resale value:** Service records valuable to buyers
- **Quality control:** Track technician performance
- **Billing accuracy:** Ensures correct charges

Service Order (Repair Order/RO) Components:

Customer Information:

- Name, address, phone number
- Email (for electronic communication)
- Alternate contact

Vehicle Information:

- Make, model, year

- VIN (Vehicle Identification Number)
- License plate number
- Current mileage/odometer reading
- Engine type (if multiple options)

Service Requested:

- What customer wants done
- Customer concerns/complaints (in their words)
- Type of service (oil change, inspection, repair)

Technician Findings:

- What was inspected
- What was found (good, wear, fail)
- Measurements (brake pad thickness, tire tread depth, etc.)
- Recommendations (urgent, soon, monitor)

Work Performed:

- What was actually done
- Parts used (with part numbers)
- Fluids added (type, quantity)
- Labor time
- Who performed work (technician name/ID)

Quality Checks:

- Test drive performed (if applicable)
- Systems verified working
- Customer concerns addressed
- Final inspection completed

Authorization:

- Customer signature (approved work)
- Technician signature (performed work)
- Advisor signature (reviewed with customer)
- Date and time

Parts Documentation:

- Part number
- Description
- Quantity

- Unit price
- Total price
- Warranty information

Labor Documentation:

- Operation description
- Labor time (hours or flat rate)
- Labor rate
- Total labor cost

Fluids Documentation:

- Type (brand, specification)
- Quantity added
- Total cost
- Proper disposal noted (used oil, coolant, etc.)

Service Sticker/Reminder:

- Date of service
- Mileage at service
- Next service due (date and mileage)
- Type of service performed
- Oil type used (5W-30, etc.)
- Technician/shop name and contact

Digital Service Records:

- Many shops now use electronic systems
- Customer email with service summary
- Photos of findings (wear, damage, leaks)
- Digital signature capture
- Automatic service reminders
- Cloud storage of history

Warranty Documentation:

- If warranty work performed
- Warranty claim number
- Authorization from warranty company
- Detailed description of failure
- Parts replaced (must use approved parts)
- Photos of failed parts

- Customer concern that led to diagnosis

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Service Schedule Application

Activity 1: Reading Owner's Manual Service Schedule (25 min)

- Students receive owner's manuals from various vehicles
- Locate service schedule section

Information to find:

- Service interval chart
- Normal vs severe service definitions
- Specific services at each interval
- Fluid specifications
- Capacities (oil, coolant, transmission)

Practice exercise:

- Given scenario: "Vehicle has 45,000 km, last service at 40,000 km"
- What services are due now?
- What services are coming soon?
- Which items are critical timing-based?

Create service plan:

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Vehicle: 2019 Maruti Swift, 45,000 km, 2.5 years old
Normal service schedule

Due NOW (40,000 km):
- Engine oil and filter change
- Engine air filter replacement
- Cabin air filter replacement
- Brake fluid replacement (2+ years old)
- Comprehensive inspection
- Tire rotation

Coming SOON (60,000 km):
- Spark plug replacement
- Drive belt inspection/replacement
- Coolant replacement (if 3+ years)
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Critical items to monitor:
- Brake pad thickness
- Tire tread depth
- Battery condition (2.5 years old)
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Activity 2: Severe vs Normal Service Determination (20 min)

- **Case studies:** Students analyze driving patterns

Example 1:

- Owner: Delivery driver in Mumbai
- Usage: 100 km/day, mostly city traffic, stop-and-go
- **Assessment:** Severe service
- **Interval adjustment:** Oil changes every 5,000 km (vs 10,000)

Example 2:

- Owner: Highway commuter
- Usage: 80 km/day, mostly highway, steady speeds
- **Assessment:** Normal service
- **Interval:** Follow standard schedule

Example 3:

- Owner: Occasional driver
- Usage: 5,000 km/year, short trips to market
- **Assessment:** Severe service (short trips + time-based aging)
- **Interval:** Every 6 months minimum (even if low mileage)

Activity 3: Creating Service Recommendations (15 min)

- Given vehicle information and inspection findings
- Students prioritize services:

Vehicle: 2018 Honda City, 72,000 km, 4 years old **Last service:** 60,000 km (1 year ago) **Findings:**

- Brake pads: 3mm front, 5mm rear
- Tires: 4mm tread depth
- Battery: 4 years old, 12.3V (engine off)
- Coolant: Never changed, dark color
- Transmission fluid: Never changed
- Small oil leak at valve cover gasket

Student task: Prioritize

- 🔴 **Urgent (Do today):**
  - Front brake pads (at minimum thickness)
  - Oil change + valve cover gasket repair
- 🟡 **Soon (Within 1 month):**
  - Coolant flush and replacement (overdue, degraded)
  - Transmission fluid change (overdue)
  - Battery replacement (old, low voltage)
- 🟢 **Monitor (Check next service):**
  - Rear brake pads (adequate life)
  - Tires (consider replacement in 10,000 km)

Pass Criteria:

- ✅ Can locate service schedule in owner's manual
- ✅ Can interpret service intervals
- ✅ Can distinguish normal vs severe service
- ✅ Can create appropriate service plan
- ✅ Can prioritize recommendations correctly

Hour 2 (2:00 PM - 3:00 PM): Multi-Point Inspection Execution

Activity 1: Comprehensive Inspection Procedure (60 min)

- **Using standardized inspection checklist**
- Students perform complete multi-point inspection

Inspection Checklist Categories:

1. Under-Hood Inspection (15 min):

- [ ] Engine oil level and condition
- [ ] Coolant level and condition
- [ ] Brake fluid level and condition
- [ ] Power steering fluid level (if equipped)
- [ ] Windshield washer fluid level
- [ ] Battery condition, terminals, hold-down
- [ ] Drive belt(s) condition and tension
- [ ] Air filter condition
- [ ] Hoses (coolant, vacuum) condition

- [ ] Visible leaks (oil, coolant, brake fluid)
- [ ] Wiring condition (chafing, damage)
- [ ] Emissions components connected

2. Under-Vehicle Inspection (15 min):

- [ ] Engine oil leaks (oil pan, valve cover)
- [ ] Transmission fluid leaks
- [ ] Differential leaks (if applicable)
- [ ] Coolant leaks (radiator, hoses, water pump)
- [ ] Brake system leaks (lines, calipers, cylinders)
- [ ] Exhaust system (leaks, damage, rust)
- [ ] Suspension components (shocks, springs, bushings)
- [ ] Steering components (tie rods, boots, rack)
- [ ] CV boots condition (if FWD/AWD)
- [ ] Driveshaft U-joints (if RWD/AWD)
- [ ] Frame/body damage or rust

3. Brake System Inspection (10 min):

- [ ] Front brake pad thickness (measure, record mm)
- [ ] Rear brake pad/shoe thickness (measure, record mm)
- [ ] Rotor condition (grooves, cracks)
- [ ] Drum condition (if applicable)
- [ ] Brake hoses condition
- [ ] Brake lines rust/damage
- [ ] Parking brake operation

4. Tire and Wheel Inspection (10 min):

- [ ] Tire tread depth (measure all 4, record mm)
- [ ] Tire wear pattern (even, center, edge, feather)
- [ ] Tire sidewall condition (cracks, bulges, damage)
- [ ] Tire pressure (all 4 + spare, adjust to spec)
- [ ] Wheel condition (bent, cracked, damaged)
- [ ] Lug nuts tight (check torque)

5. Lighting and Electrical (5 min):

- [ ] Headlights (low and high beam)
- [ ] Fog lights (if equipped)
- [ ] Turn signals (all 4 corners + side markers)
- [ ] Brake lights (all)

- [ ] Reverse lights
- [ ] License plate light
- [ ] Interior lights
- [ ] Dashboard warning lights (check during start)
- [ ] Horn operation
- [ ] Wiper operation (all speeds)
- [ ] Washer spray

6. Interior and Safety (5 min):

- [ ] Seat belts (all positions, operation, condition)
- [ ] SRS warning light (check operation)
- [ ] Airbag covers (intact, no damage)
- [ ] Horn operation
- [ ] Mirrors (adjust properly)
- [ ] HVAC operation (heat, A/C, fan speeds)
- [ ] Power windows operation
- [ ] Power locks operation
- [ ] Instrument cluster (all gauges work)

Students complete full inspection:

- Work systematically through checklist
- Record measurements (brake pads, tires, fluids)
- Take photos of any issues
- Note condition: ✅ Good / ⚠️ Monitor / ❌ Service Needed

Pass Criteria:

- ✅ Completes all inspection points
- ✅ Records measurements accurately
- ✅ Documents findings clearly
- ✅ Takes supporting photos
- ✅ Works systematically (doesn't skip items)

Hour 3 (3:00 PM - 4:00 PM): PM Execution & Documentation

Activity 1: Oil Change Service Execution (30 min)

- Students perform complete oil change service
- **Following professional procedure:**

Preparation:

1. Verify correct oil type and quantity (owner's manual)
2. Verify correct filter part number
3. Gather tools (wrenches, drain pan, funnel, rags)
4. Safety check (vehicle stable, parking brake, chocks)

Execution:

1. Warm engine (5 min) - hot oil drains faster
2. Lift vehicle safely
3. Locate drain plug and filter
4. Position drain pan
5. Remove drain plug (last few turns by hand)
6. Let drain completely (5-10 min)
7. Inspect drain plug and washer
8. Clean drain plug threads
9. Install NEW crush washer (if required)
10. Reinstall drain plug - torque to spec (25-30 Nm typical)
11. Remove old filter (oil will drain)
12. Clean filter mounting surface
13. Apply thin oil film to new filter gasket
14. Install new filter (hand tight + 3/4 turn)
15. Lower vehicle
16. Remove oil filler cap
17. Add specified quantity of oil (funnel)
18. Install filler cap
19. Start engine, run 1 minute
20. Check for leaks (drain plug, filter)
21. Shut off engine, wait 2 minutes
22. Check oil level with dipstick
23. Add to FULL mark if needed
24. Recheck for leaks

Quality Checks:

- No leaks at drain plug
- No leaks at filter
- Oil level at FULL mark
- Oil cap installed
- Tools removed from engine bay
- Oil spills cleaned up

Activity 2: Service Documentation Completion (20 min)

- **Complete service order form:**

Student fills out:

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SERVICE ORDER

Customer: [Name]
Vehicle: 2019 Maruti Swift VIN: [VIN]
Mileage: 45,280 km
Date: [Today's date]

SERVICES REQUESTED:
- 40,000 km service per schedule

TECHNICIAN FINDINGS:
Fluids:
- Engine oil: Dark, due for change ✅
- Coolant: Good level, clean ✅
- Brake fluid: Good level ✅
- Washer fluid: Low ⚠️

Tires:
- LF: 6mm tread, 32 PSI ✅
- RF: 6mm tread, 30 PSI (adjusted to 32) ✅
- LR: 5mm tread, 30 PSI (adjusted to 32) ✅
- RR: 5mm tread, 31 PSI (adjusted to 32) ✅

Brakes:
- Front pads: 5mm remaining ✅
- Rear pads: 7mm remaining ✅
- Rotors: Good condition ✅

Filters:
- Engine air filter: Dirty ⚠️ (replaced)
- Cabin air filter: Dirty ⚠️ (replaced)

WORK PERFORMED:
- Engine oil change: 3.2L 5W-30 synthetic
- Oil filter replaced: Part# 15400-RTA-003
- Engine air filter replaced: Part# 17220-RNA-A00
- Cabin air filter replaced: Part# 80292-SDG-W01
- Tire pressures adjusted to 32 PSI
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- Washer fluid topped up
- Multi-point inspection completed
- No leaks found
- Test drive: All systems functioning

RECOMMENDATIONS:
🟢 Monitor (10,000 km):
- Brake pads: Adequate remaining life
- Tires: Good tread depth

PARTS USED:
- Engine oil, 5W-30 synthetic: 3.2L @ ₹450/L = ₹1,440
- Oil filter Part# 15400-RTA-003: 1 @ ₹250 = ₹250
- Air filter Part# 17220-RNA-A00: 1 @ ₹350 = ₹350
- Cabin filter Part# 80292-SDG-W01: 1 @ ₹450 = ₹450
- Washer fluid: 0.5L @ ₹100/L = ₹50

LABOR:
- Oil change service: 0.5 hrs @ ₹800/hr = ₹400
- Air filter replacement: 0.2 hrs @ ₹800/hr = ₹160
- Multi-point inspection: Included

TOTAL PARTS: ₹2,540
TOTAL LABOR: ₹560
SUBTOTAL: ₹3,100
TAX (18% GST): ₹558
TOTAL: ₹3,658

Next service due: 50,000 km or 6 months (whichever first)

Technician: _____ Date: _____
Customer Signature: _____ Date: _____
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Activity 3: Service Sticker Completion (10 min)

- Fill out service reminder sticker:
  - Date of service
  - Mileage at service
  - Next service due date
  - Next service due mileage
  - Oil type used
  - Shop contact info
- Place sticker (windshield or door jamb)

Pass Criteria:

- ✅ Performs oil change correctly
- ✅ Torques drain plug to specification
- ✅ Checks for leaks
- ✅ Completes service documentation fully
- ✅ Records all parts and labor
- ✅ Calculates costs correctly
- ✅ Creates professional service sticker

DAY 25 ASSESSMENT:

Practical Skills Checklist:

- [ ] Reads service schedule from owner's manual
- [ ] Determines normal vs severe service requirements
- [ ] Creates appropriate service plan
- [ ] Performs multi-point inspection systematically
- [ ] Records measurements accurately
- [ ] Executes oil change service properly
- [ ] Completes service documentation
- [ ] Calculates service costs correctly
- [ ] Prioritizes recommendations appropriately

Knowledge Check (Verbal):

- What is the difference between normal and severe service?
- Give three examples of severe service conditions
- Why is timing belt replacement critical?
- What should be documented on a service order?
- Why do we need service stickers?

DAY 26: QUALITY STANDARDS + PROFESSIONAL DOCUMENTATION

Learning Objectives:

- Understand workshop quality standards (5S, safety)
- Apply torque specifications correctly

- Create professional documentation
- Take effective photos for customers
- Understand warranty procedures

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Workshop Quality Standards

5S Methodology:

- **Japanese workplace organization system**
- **Improves efficiency, safety, quality**

1. Seiri (Sort):

- **Separate needed from unneeded**
- Remove unnecessary items from workspace
- Keep only what's needed for current job
- **Benefits:**
  - Less clutter
  - Find tools faster
  - Safer work environment
- **Practice:**
  - Red tag items not used in 30 days
  - Remove or relocate
  - Keep work area clean

2. Seiton (Set in Order):

- **"A place for everything, everything in its place"**
- Organize tools and materials logically
- Most-used items most accessible
- Label storage locations
- **Benefits:**
  - No time wasted searching
  - Easy to see what's missing
  - Standardized locations
- **Practice:**
  - Shadow boards for tools
  - Labeled drawers
  - Color coding

    - Tool inventory lists

3. Seiso (Shine):

- **Keep workspace clean**
- Clean as you work, not just at end of day
- Cleaning = inspection opportunity
- **Benefits:**
    - Spot problems early (leaks, damage)
    - Professional appearance
    - Equipment lasts longer
- **Practice:**
    - Daily workspace cleaning
    - Wipe tools after use
    - Clean spills immediately
    - Preventive cleaning schedule

4. Seiketsu (Standardize):

- **Create standards for first 3 S's**
- Make it routine
- Checklists and procedures
- **Benefits:**
    - Consistent quality
    - New employees learn faster
    - Easier to maintain
- **Practice:**
    - Standard work procedures
    - Cleaning schedules
    - Inspection checklists
    - Visual management

5. Shitsuke (Sustain):

- **Maintain discipline**
- Make 5S a habit
- Continuous improvement
- **Benefits:**
    - Long-term results
    - Culture of quality
    - Pride in workspace
- **Practice:**

  - Regular audits
  - Team accountability
  - Recognition for good practices
  - Training reinforcement

Workshop Safety as Quality Standard:

- **Clean floors** (no oil, no trip hazards)
- **Proper lighting** (see clearly)
- **Fire extinguisher** (accessible, inspected)
- **First aid kit** (stocked, accessible)
- **Safety equipment** (glasses, gloves available)
- **Lifting equipment** (inspected, certified)
- **Ventilation** (exhaust fumes removed)
- **Electrical safety** (no damaged cords)
- **Chemical storage** (labeled, secured, MSDS available)

Professional Work Standards:

- **Cleanliness:** Vehicle returned cleaner than received
- **Protection:** Seat covers, floor mats, fender covers
- **Thoroughness:** Job done completely, not half-done
- **Double-checking:** Verify work before returning vehicle
- **Test drive:** Ensure customer concern resolved
- **Final inspection:** Nothing left undone

Hour 2 (10:00 AM - 11:00 AM): Torque Specifications

Why Torque Matters:

- **Too loose:** Part can fall off, leak, fail
- **Too tight:** Strip threads, break bolts, warp parts
- **Just right:** Proper clamping force, seal, reliability

Understanding Torque:

- **Torque = Rotational force**
- **Units:**
  - Newton-meters (Nm) - metric, most common
  - Foot-pounds (ft-lbs) - imperial
  - Kilogram-meters (kgm) - older metric

- **Example:** 85 Nm = 63 ft-lbs = 8.7 kgm

Torque Conversion (Awareness):

- 1 Nm = 0.74 ft-lbs
- 1 ft-lb = 1.36 Nm
- Most torque wrenches have dual scales

Why Specifications Vary:

- **Thread size:** Larger bolts = more torque
- **Material:** Steel vs aluminum
- **Application:** Critical safety vs non-critical
- **With/without lubricant:** Lubricated = less torque needed

Common Torque Specifications:

Wheel Lug Nuts:

- **Small cars (steel wheels):** 80-100 Nm
- **Mid-size cars (alloy wheels):** 100-120 Nm
- **SUVs/trucks:** 120-140 Nm
- **Always check vehicle-specific spec!**

Oil Drain Plug:

- **Typical:** 25-35 Nm
- **Over-tightening strips threads in oil pan**
- **Aluminum pans especially vulnerable**

Spark Plugs:

- **14mm thread:** 20-30 Nm
- **18mm thread:** 30-40 Nm
- **Torque critical - too loose = blow out, too tight = break**

Cylinder Head Bolts:

- **50-100+ Nm typically**
- **Often require sequence AND multiple passes**
- **Some require angle torque (degrees)**
- **Very critical - affects gasket seal**

Suspension/Steering:

- **Ball joints:** 50-80 Nm

- **Tie rod ends:** 40-60 Nm
- **Control arm bolts:** 80-120 Nm
- **Safety critical!**

Caliper Bolts:

- **Caliper bracket:** 80-120 Nm
- **Caliper slide pins:** 25-35 Nm
- **Bleeder screw:** 8-12 Nm

Torque Sequences:

- **Some fasteners must be torqued in specific order**
- **Cylinder head bolts:** Spiral pattern from center out
- **Oil pan bolts:** Crisscross pattern
- **Wheel lug nuts:** Star pattern
- **Why?** Distribute clamping force evenly, prevent warping

Multi-Stage Torque:

- **Some fasteners require multiple passes**
- **Example: Cylinder head**
  1. First pass: 30 Nm (all bolts)
  2. Second pass: 60 Nm (all bolts)
  3. Third pass: 90 Nm (all bolts)
  4. Fourth pass: +90° (angle torque)
- **Why?** Ensures even clamping, prevents distortion

Angle Torque (Torque-to-Yield):

- **Some modern fasteners use angle instead of torque**
- **Torque to initial value, then rotate additional degrees**
- **Example:** 40 Nm + 90°
- **Requires angle gauge**
- **Bolts usually one-time use (don't reuse)**

Hour 3 (11:00 AM - 12:00 PM): Documentation Best Practices

Why Documentation Quality Matters:

- **Legal protection** for shop and customer
- **Professional image** builds trust

- **Warranty compliance** requires proper records
- **Future reference** helps diagnosis
- **Billing accuracy** prevents disputes
- **Quality control** tracks performance

Elements of Professional Documentation:

1. Clarity:

- Write legibly (or type)
- Use proper terminology
- Avoid abbreviations unless standard
- Be specific, not vague
  - ❌ "Fixed noise"
  - ✅ "Replaced worn left front wheel bearing, noise eliminated"

2. Completeness:

- Record everything relevant
- Measurements with units
- Part numbers
- Before and after conditions
- Who performed work
- Date and time

3. Accuracy:

- Record actual findings
- Don't embellish or minimize
- Measurements must be precise
- Double-check numbers (mileage, costs)

4. Objectivity:

- State facts, not opinions
- "Tire tread 3mm" not "tire worn out" (customer decides)
- Use measurements, not adjectives
  - ❌ "Bad brake pads"
  - ✅ "Brake pads 2mm thickness, minimum 3mm"

Photo Documentation:

- **Becoming industry standard**
- **Shows customer exactly what technician sees**

- **Eliminates miscommunication**

What to Photograph:

- Worn parts (brake pads, tires)
- Damage (body, glass, existing scratches)
- Leaks (fluid puddles, wet spots)
- Rust/corrosion
- Measurements (ruler/gauge visible)
- Failed parts removed
- Quality of repair

Photo Best Practices:

- Good lighting (use flashlight if needed)
- In focus
- Include reference (ruler, coin for scale)
- Multiple angles if needed
- Label/annotate photos
- Store with service order

Digital Inspection Apps:

- Many shops now use tablets/phones
- Take photos directly into service order
- Annotate with arrows and notes
- Email report to customer immediately
- Customer can approve work remotely

Warranty Documentation Requirements:

- **More detailed than regular service**
- Must prove failure was defect, not abuse
- Photos of failed part essential
- Description of failure mode
- Related codes/diagnostics
- Parts must be genuine/approved
- Customer authorization
- Submission deadlines (usually 30 days)

Record Retention:

- **Service orders:** Keep 5+ years (legal requirement varies)
- **Warranty claims:** Keep until claim paid + 2 years

- **Customer database:** Indefinitely (for service history)
- **Digital preferred** (easier to search, doesn't degrade)

Data Privacy:

- **Customer information is confidential**
- Don't share customer details
- Secure storage (locked files, password-protected digital)
- Comply with data protection regulations
- Only authorized access

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): 5S Implementation Practice

Activity 1: Workspace Organization Exercise (30 min)

- **Scenario:** Messy workbench provided
- Students apply 5S principles:

1. Sort (Seiri) - 10 min:

- Identify needed tools for current job (oil change)
- Separate unneeded items (other tools, parts, trash)
- Create "keep" and "remove" piles
- Remove all non-essential items

2. Set in Order (Seiton) - 10 min:

- Arrange needed tools logically:
    - Wrenches by size
    - Drain pan accessible
    - Funnel near oil
    - Rags available
- Create efficient workspace layout
- Label locations (if permanent)

3. Shine (Seiso) - 5 min:

- Clean workbench surface
- Wipe tools
- Sweep floor around workspace
- Inspect for damage/wear

4. Standardize (Seiketsu) - 5 min:

- Create simple procedure:
  - "Before starting any job: Sort, Organize, Clean"
  - End-of-day checklist
- Take "after" photo of organized workspace

Students compare before/after:

- Time to find tools (before: 2 min, after: 10 seconds)
- Safety (before: trip hazards, after: clear floor)
- Professional appearance

Activity 2: Tool Accountability System (15 min)

- **Shadow board demonstration:**
  - Each tool has designated spot
  - Tool outline drawn/labeled
  - Missing tool immediately visible

Students create mini shadow board:

- Select 10 common tools
- Trace outlines on cardboard
- Label each position
- Place tools in positions
- Remove one tool (simulate use)
- Verify easy to see what's missing

Activity 3: End-of-Day Cleanup Procedure (15 min)

- **Develop and execute cleanup checklist:**

End-of-Day Checklist:

- [ ] Return all tools to designated storage
- [ ] Clean all tools used
- [ ] Sweep floor (no oil, debris)
- [ ] Empty trash containers
- [ ] Return parts to storage
- [ ] Wipe down workbench
- [ ] Turn off equipment
- [ ] Check for forgotten items under vehicle
- [ ] Secure shop (if last to leave)

- [ ] Report any damaged tools/equipment

Students execute cleanup:

- Work systematically through list
- Time themselves (should be 10-15 min)
- Instructor inspects for completeness

Pass Criteria:

- ✅ Can apply 5S principles effectively
- ✅ Creates organized workspace
- ✅ Demonstrates tool accountability system
- ✅ Executes thorough end-of-day cleanup
- ✅ Works systematically, not randomly

Hour 2 (2:00 PM - 3:00 PM): Torque Specification Application

Activity 1: Torque Wrench Setup & Use Practice (25 min)

- **Review torque wrench types:**
  - Click-type (most common)
  - Beam-type (simple, reliable)
  - Digital (expensive, precise)

Setting click-type torque wrench:

1. Check calibration status (should be recent)
2. Unlock handle (usually rotate collar)
3. Adjust to desired torque value
   - Read main scale
   - Read vernier scale
   - Total = both combined
4. Lock handle
5. Apply smoothly to fastener
6. Listen/feel for click
7. **STOP immediately at click** (don't continue pulling)

Students practice setting torque wrench:

- Set to 25 Nm
- Set to 85 Nm

- Set to 120 Nm
- Verify correct reading each time

Activity 2: Lug Nut Torque Practice (25 min)

- **Real-world application: Wheel installation**

Procedure:

1. Look up specification: 105 Nm (example vehicle)
2. Set torque wrench to 105 Nm
3. Install wheel (review from Day 1, 13)
4. Start all lug nuts by hand
5. Snug in star pattern with regular wrench
6. Lower vehicle (tire touching ground)
7. **Final torque in star pattern:**
   - Lug nut 1 → click
   - Lug nut 3 (opposite) → click
   - Lug nut 5 → click
   - Lug nut 2 → click
   - Lug nut 4 → click
8. Go around again to verify all torqued
9. Return torque wrench to lowest setting
10. Store properly

Common mistakes to avoid:

- Using impact wrench for final tightening (can over-torque)
- Pulling past click
- Not using star pattern
- Wrong torque specification
- Not storing wrench at lowest setting

Students torque wheels on multiple vehicles:

- Different specifications for each
- Must look up spec for each vehicle
- Demonstrates importance of verification

Activity 3: Oil Drain Plug Torque Practice (10 min)

- **Small fastener, critical not to over-tighten**

Procedure:

1. Look up specification: 29 Nm (example)
2. Set torque wrench to 29 Nm
3. Install new crush washer
4. Install drain plug by hand (ensure threads start correctly)
5. Tighten with torque wrench until click
6. Verify no cross-threading

Students practice:

- Feel the difference between 29 Nm (drain plug) and 105 Nm (lug nut)
- Understand that small fasteners need less torque
- Appreciate risk of stripping aluminum oil pans

Pass Criteria:

- ✅ Can set torque wrench correctly to any value
- ✅ Applies torque properly (smooth pull, stops at click)
- ✅ Uses correct sequence (star pattern for wheels)
- ✅ Looks up specifications before torquing
- ✅ Stores torque wrench properly (lowest setting)

Hour 3 (3:00 PM - 4:00 PM): Professional Documentation Practice

Activity 1: Photo Documentation Exercise (25 min)

- **Students document findings with photos**

Scenario: Pre-purchase inspection findings

- Worn brake pads
- Low tire tread
- Oil leak
- Damaged windshield
- Rust on exhaust

For each finding:

1. Take clear, well-lit photo
2. Include measurement tool in photo (if applicable)
3. Take multiple angles if needed
4. Annotate photo with description
5. Rate severity (good/monitor/replace)

Photo quality checklist:

- [ ] In focus
- [ ] Good lighting (use flashlight if needed)
- [ ] Shows problem clearly
- [ ] Includes context (not too close)
- [ ] Measurement visible (brake pad thickness, tire tread)
- [ ] Clean lens
- [ ] Horizontal orientation (not tilted)

Students review each other's photos:

- Can you see the problem clearly?
- Would customer understand from photo alone?
- Is measurement visible and readable?
- Does photo support the finding?

Activity 2: Complete Service Documentation (25 min)

- **Given scenario and findings, create professional documentation**

Scenario:

- Vehicle: 2020 Hyundai Creta, 58,000 km
- Customer concern: "Squeaking noise from brakes"
- Inspection findings:
  - Front brake pads: 2mm (minimum 3mm)
  - Rear brake pads: 6mm
  - Front rotors: Grooved, within thickness spec
  - Brake fluid: Dark, 3 years old
  - Tires: 5mm tread, uneven wear on front
  - Battery: 12.4V, 3 years old, terminals corroded
  - Oil: Clean, recent service evident

Students create complete service order:

Customer concern (in their words): "Squeaking noise when braking, especially in the morning"

Technician findings:

```
BRAKE INSPECTION:
- Front brake pads: 2mm thickness (BELOW 3mm minimum) ❌
  - Wear indicator contacting rotor (source of squeak)
```

```
  - Photo attached
- Rear brake pads: 6mm thickness ✅
- Front rotors: Grooved from worn pads ⚠️
  - Measured thickness: 25.3mm (minimum 24.0mm)
  - Within spec but recommend resurfacing or replacement
- Brake fluid: Dark brown color, 3+ years old ⚠️
  - Recommend replacement (hygroscopic, performance degraded)

TIRE INSPECTION:
- Front left: 5mm tread, inside edge wear ⚠️
- Front right: 5mm tread, inside edge wear ⚠️
- Rear left: 6mm tread, even wear ✅
- Rear right: 6mm tread, even wear ✅
- Uneven front wear indicates alignment issue

BATTERY:
- Voltage: 12.4V (marginal) ⚠️
- Age: 3 years (expected life 4-5 years)
- Terminals: Heavy corrosion ⚠️
- Load test: Recommend

OTHER:
- Engine oil: Clean, adequate level ✅
- Coolant: Good level, clean ✅
- All lights functional ✅
```

Recommendations:

```
🔴 URGENT (Safety/Immediate):
1. Replace front brake pads: ₹2,500 parts + ₹800 labor = ₹3,300
   - Squeaking noise will stop
   - Safety critical - pads at minimum

🟡 RECOMMENDED (Soon - within 1 month):
2. Resurface or replace front rotors: ₹3,000-4,500
   - Grooved from worn pads
   - Better braking performance
3. Brake fluid flush and replacement: ₹1,200
   - 3+ years old, degraded
   - Better pedal feel and performance
4. Clean battery terminals: ₹200
   - Improve electrical connection
   - Prevent starting issues
```

```
5. Wheel alignment: ₹1,500
   - Uneven tire wear indicates misalignment
   - Extend tire life

🟢 MONITOR (Next service in 10,000 km):
6. Rear brake pads: Adequate life remaining
7. Battery: Monitor voltage, consider replacement in 1 year
8. Tires: 5mm tread adequate for now, monitor wear pattern after alignment

TOTAL FOR URGENT WORK: ₹3,300 + tax
TOTAL FOR ALL RECOMMENDED: ₹9,400 + tax
```

Customer authorization: "Authorize urgent brake pad replacement today. Will schedule alignment next month." Signature: _

Activity 3: Documentation Review & Feedback (10 min)

- Students exchange documentation
- Peer review using checklist:
  - ☐ Clear description of problem
  - ☐ Measurements recorded with units
  - ☐ Recommendations prioritized correctly
  - ☐ Costs calculated accurately
  - ☐ Professional language used
  - ☐ Photos support findings
  - ☐ Customer can understand without technical knowledge
- Instructor provides feedback on best/worst examples

Pass Criteria:

- ✅ Takes clear, useful photos
- ✅ Includes measurements in photos
- ✅ Creates complete, professional documentation
- ✅ Prioritizes recommendations appropriately
- ✅ Uses clear, professional language
- ✅ Calculates costs accurately
- ✅ Work is legible and organized

DAY 26 ASSESSMENT:

Practical Skills Checklist:

- [ ] Applies 5S principles to workspace
- [ ] Creates organized, safe work environment
- [ ] Sets torque wrench correctly
- [ ] Applies torque properly (stops at click)
- [ ] Uses correct torque sequence
- [ ] Looks up specifications before torquing
- [ ] Takes clear, useful documentation photos
- [ ] Creates professional service documentation
- [ ] Prioritizes recommendations correctly
- [ ] Uses professional language

Knowledge Check (Verbal):

- What do the 5 S's stand for?
- Why is torque important?
- What happens if you over-tighten lug nuts?
- What should be included in photo documentation?
- Why do we prioritize recommendations as urgent/soon/monitor?

DAY 27: TOOL MANAGEMENT + WORKSHOP IT SYSTEMS

Learning Objectives:

- Implement personal tool inventory management
- Understand tool maintenance and care procedures
- Navigate parts catalog software
- Use VIN decoding for parts identification
- Access online service information systems

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Tool Management Fundamentals

Why Tool Management Matters:

- **Tools are expensive** (₹50,000-200,000 initial investment)
- **Missing tools = lost productivity**
- **Damaged tools = safety risk**
- **Professional image** requires proper tools

- **Career investment** - tools last entire career if maintained

Personal Tool Ownership:

- **Most technicians own their tools**
- **Shop provides:**
  - Lifts and large equipment
  - Specialty tools (expensive, rarely used)
  - Diagnostic equipment
  - Air compressor, power tools
- **Technician provides:**
  - Hand tools (wrenches, sockets, screwdrivers)
  - Basic power tools (impact wrench, drill)
  - Measuring tools (calipers, micrometers)
  - Personal toolbox

Tool Investment Strategy:

- **Start with basics:** Essential tools first
- **Buy quality:** Lifetime warranty brands worth it
- **Add gradually:** Build collection over time
- **Avoid cheap tools:** Break easily, dangerous
- **Tool brands:**
  - **Professional:** Snap-on, Mac, Matco (expensive, lifetime warranty)
  - **Mid-range:** Stanley, Craftsman, Sidchrome (good quality, affordable)
  - **Budget:** Indian brands (acceptable for non-critical use)

Tool Inventory Management:

- **Complete inventory list:**
  - Every tool documented
  - Description, size, quantity
  - Purchase date and cost
  - Serial numbers (if applicable)
  - Location (which box, drawer)
- **Why inventory needed:**
  - Insurance (theft, fire)
  - Tax purposes (business expense)
  - Track missing tools
  - Plan future purchases

Tool Organization Systems:

1. Tool Box Organization:

- **Top box:** Frequently used tools
  - Screwdrivers, pliers, common wrenches
  - Easy access without bending
- **Middle drawers:** Organized by type
  - Drawer 1: Sockets (metric)
  - Drawer 2: Sockets (imperial) if needed
  - Drawer 3: Wrenches
  - Drawer 4: Specialty tools
- **Bottom drawers:** Heavy or less-used items
  - Hammers, pry bars
  - Specialty sockets
  - Rarely-used tools

2. Socket Organization:

- **Organized by size** (small to large)
- **Separate drive sizes:**
  - 1/4" drive (small fasteners)
  - 3/8" drive (general use)
  - 1/2" drive (heavy fasteners)
  - 3/4" drive (very heavy - wheels, suspension)
- **Socket rails or trays:**
  - Each socket has designated spot
  - Size marked clearly
  - Missing socket immediately obvious

3. Wrench Organization:

- **Combination wrenches:** Size order (8mm, 10mm, 12mm, etc.)
- **Wrench racks:** Clip or magnetic
- **Separate metric and imperial** (if both used)

4. Screwdriver Organization:

- **By type:** Phillips, flat, Torx, etc.
- **Then by size:** Small to large
- **Standing holders** or **wall-mounted racks**

Tool Accountability:

- **Shadow boards:** Tool outlines show missing items

- **Tool check-in/check-out** (for shop tools)
- **Daily inventory** of critical/expensive tools
- **End-of-job checks:** All tools accounted for before moving to next job
- **Never leave tools in/under vehicle** when returning to customer

---

Hour 2 (10:00 AM - 11:00 AM): Tool Maintenance & Calibration

Tool Maintenance Procedures:

Hand Tools (Daily/Weekly):

- **Cleaning:**
  - Wipe after use (remove oil, grease, dirt)
  - Prevent rust (especially in humid climates)
  - Use parts cleaner for heavy contamination
  - Dry thoroughly
- **Inspection:**
  - Cracks (especially wrenches - discard if cracked)
  - Wear (socket corners rounded - replace)
  - Damage (bent, broken)
  - Corrosion (light rust ok, heavy = replace)
- **Lubrication:**
  - Moving parts (ratchet mechanisms, pliers)
  - Light oil only (WD-40, machine oil)
  - Don't over-lubricate (attracts dirt)

Power Tools:

- **Air tools:**
  - Daily: Add pneumatic tool oil (few drops)
  - Weekly: Clean air filter
  - Check air pressure (90 PSI typical)
  - Inspect hoses for damage
- **Electric tools:**
  - Keep clean and dry
  - Inspect cords for damage
  - Check brushes periodically
  - Store batteries charged (cordless)

Cutting Tools:

- **Files:** Clean with wire brush (card)
- **Hacksaw blades:** Replace when dull
- **Drill bits:** Sharpen or replace when dull
- **Taps and dies:** Clean threads, light oil

Measuring Tools (Critical Maintenance):

- **Calipers:**
  - Never drop (affects calibration)
  - Close jaws when storing
  - Light oil on sliding surfaces
  - Clean with alcohol (not water)
  - Verify zero before use
- **Micrometers:**
  - Store in case
  - Clean spindle and anvil
  - Check calibration with gauge block
  - Never force closed
  - Light oil on threads
- **Dial indicators:**
  - Protect glass/face
  - Store in case
  - Check for smooth movement
  - Verify calibration periodically

Torque Wrench Maintenance (CRITICAL):

- **After each use:**
  - Return to lowest setting (releases spring tension)
  - Wipe clean
  - Store horizontally in case
- **Never:**
  - Use as breaker bar (will damage calibration)
  - Drop or strike
  - Over-torque past maximum
  - Store at high setting
- **Calibration:**
  - Check annually (or per manufacturer)
  - Professional calibration service
  - Calibration certificate for warranty work
  - Replace if consistently off

Calibration Awareness:

- **What is calibration?**
  - Verification that tool measures accurately
  - Comparison to known standard
  - Adjustment if needed
  - Documentation of accuracy

Tools requiring calibration:

- Torque wrenches (most critical)
- Micrometers
- Calipers
- Pressure gauges
- Multimeters (advanced work)
- Scan tools (software updates)

Calibration frequency:

- **Torque wrenches:** Annually or 5,000 uses
- **Measuring tools:** Annually or if dropped
- **Pressure gauges:** Annually
- **More frequent if:**
  - Heavy use
  - Critical applications (engine building)
  - Warranty work requirements

Calibration services:

- Professional calibration labs
- Tool manufacturer services
- Certificate of calibration provided
- Sticker on tool showing calibration date

Tool Storage:

- **Protect from:**
  - Moisture (causes rust)
  - Dust (damages precision tools)
  - Impact (dropping, banging)
  - Theft (lock toolbox)
- **Storage conditions:**
  - Dry environment

  - Moderate temperature
  - Secure (toolbox locks)
  - Organized (prevents damage from tools hitting each other)

Tool Safety:

- **Damaged tools are dangerous:**
  - Cracked wrench can break under load
  - Rounded socket slips, damages fastener
  - Damaged screwdriver slips, causes injury
- **Remove damaged tools from service immediately**
- **Mark clearly "DAMAGED - DO NOT USE"**
- **Repair or replace, don't ignore**

Hour 3 (11:00 AM - 12:00 PM): Workshop IT Systems

Digital Tools in Modern Workshops:

- **Replacing paper manuals and catalogs**
- **Faster access to information**
- **Always up-to-date**
- **Essential for modern vehicle service**

Parts Catalog Software:

Purpose:

- Identify correct part for specific vehicle
- Get accurate part numbers
- Check parts availability
- View exploded diagrams
- Verify compatibility

How to use:

1. **Enter vehicle identification:**
   - VIN (most accurate)
   - Or Make/Model/Year/Engine
2. **Select system/component:**
   - Navigate menus (Engine, Transmission, Brakes, etc.)
   - Drill down to specific component
3. **View parts diagram:**

- Exploded view shows all parts
- Each part numbered
- Click number for details

4. **Get part information:**
   - Part number
   - Description
   - Price
   - Availability
   - Superseded part numbers (updates)
5. **Order parts:**
   - Add to cart
   - Submit order
   - Track delivery

Common parts catalog systems:

- **Manufacturer portals:** Maruti, Hyundai, Tata, etc.
- **Aftermarket:** AUTODOC, Partsfinder, etc.
- **Local distributors:** Region-specific systems

VIN Decoding:

- **VIN = Vehicle Identification Number**
- **17 characters (letters and numbers)**
- **Unique to each vehicle**

VIN structure:

```
Example: MA3ERLF1S00289764

Position 1-3: World Manufacturer Identifier (WMI)
  MA3 = Maruti Suzuki India

Position 4-8: Vehicle Descriptor Section (VDS)
  ERLF1 = Model, body type, engine type, restraint system

Position 9: Check digit (validation)
  S = Check digit

Position 10: Model year
  0 = 2000 (or 2010, 2020 - cycles every 30 years)

Position 11: Plant code
```

```
  0 = Assembly plant

Position 12-17: Sequential production number
  289764 = Unique serial number
```

Using VIN for parts:

- Enter VIN in catalog
- System automatically identifies:
  - Exact model and trim
  - Engine type
  - Transmission type
  - Optional equipment
  - Build date
- Ensures correct parts ordered

Where to find VIN:

- Dashboard (visible through windshield, driver side)
- Driver door jamb (sticker or plate)
- Registration documents
- Insurance papers
- Engine bay (firewall or strut tower)

Online Service Information Systems:

Types of service information:

- **Repair procedures:** Step-by-step instructions
- **Wiring diagrams:** Electrical troubleshooting
- **Specifications:** Torque values, capacities, clearances
- **TSBs (Technical Service Bulletins):** Known issues and fixes
- **Recall information:** Safety recalls
- **Maintenance schedules:** Service intervals

Information sources:

- **Manufacturer websites:**
  - Official repair manuals
  - Most accurate
  - May require subscription
  - Example: Maruti TechInfo, Hyundai GDS
- **Aftermarket platforms:**

    - Mitchell1, AllData, Haynes Pro
    - Cover multiple brands
    - Subscription-based
    - Used by independent shops
- **Free resources:**
    - Owner's manuals online
    - YouTube (use with caution - verify accuracy)
    - Forums (technician communities)

Navigating service information:

1. Select vehicle (VIN or details)
2. Choose category:
    - Maintenance
    - Engine
    - Transmission
    - Brakes
    - Electrical
    - etc.
3. Select topic:
    - Removal/Installation
    - Diagnosis
    - Specifications
    - Wiring
4. Read procedure
5. Print or save for reference

Workshop Management Software:

Functions:

- **Customer database:** Contact info, vehicle history
- **Appointment scheduling:** Book services, track capacity
- **Work orders:** Create, track, close
- **Inventory management:** Parts stock, ordering
- **Invoicing:** Generate bills, track payments
- **Reports:** Sales, productivity, profitability

Benefits:

- Paperless (or less paper)
- Faster service

- Better customer communication
- Track vehicle history
- Improve efficiency
- Data analysis for business decisions

Digital Inspection Apps:

- Tablet/phone-based
- Take photos during inspection
- Annotate findings
- Email report to customer
- Customer can approve work remotely
- Increases transparency and trust

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Tool Inventory & Organization

Activity 1: Personal Tool Inventory Creation (30 min)

- Students create inventory of their tools (or sample toolset)

Inventory format:

```
TOOL INVENTORY – [Student Name]
Date: [Today's date]

WRENCHES – Combination (Metric)
– 8mm combination wrench | Qty: 1 | Brand: Stanley | Purchase: 2024 |
Cost: ₹150
– 10mm combination wrench | Qty: 1 | Brand: Stanley | Purchase: 2024 |
Cost: ₹180
– 12mm combination wrench | Qty: 1 | Brand: Stanley | Purchase: 2024 |
Cost: ₹200
[continue for all sizes...]

SOCKETS – 1/2" Drive (Metric)
– 10mm deep socket | Qty: 1 | Brand: Taparia | Purchase: 2024 | Cost: ₹80
– 12mm deep socket | Qty: 1 | Brand: Taparia | Purchase: 2024 | Cost: ₹90
[continue...]

SCREWDRIVERS
```

```
- Phillips #2 screwdriver | Qty: 1 | Brand: Taparia | Purchase: 2024 |
Cost: ₹120
- Flat 6mm screwdriver | Qty: 1 | Brand: Taparia | Purchase: 2024 | Cost:
₹100
[continue...]

SPECIALTY TOOLS
- Digital caliper 150mm | Qty: 1 | Brand: Mitutoyo | Purchase: 2024 |
Cost: ₹2,500 | Serial: 12345
- Torque wrench 40-200Nm | Qty: 1 | Brand: Norbar | Purchase: 2024 | Cost:
₹8,500 | Calibration: Jan 2025

TOTAL INVESTMENT: ₹45,680
```

Digital inventory:

- Use spreadsheet (Excel, Google Sheets)
- Add photo of each major tool
- Update when tools added
- Note if tool loaned out
- Insurance purposes

Activity 2: Tool Organization Practice (30 min)

- **Given disorganized toolbox**
- Students organize using best practices:

Organization steps:

1. **Empty entire toolbox**
2. **Clean toolbox drawers**
3. **Sort tools by category:**
   - Wrenches
   - Sockets (by drive size)
   - Screwdrivers
   - Pliers
   - Specialty tools
4. **Assign drawer/location:**
   - Most-used tools in top drawer
   - Heavy items in bottom
5. **Use organizers:**
   - Socket rails
   - Wrench racks

    - Divided trays
6. **Create labels** (if permanent)
7. **Document organization** (photo for reference)

Time tracking:

- Initial organization: 30 minutes
- Daily retrieval of random tool: 10 seconds
- End-of-day return all tools: 5 minutes

Comparison:

- **Organized:** Find any tool in 10 seconds
- **Disorganized:** 2-5 minutes searching (lost productivity)

Pass Criteria:

- ✅ Creates complete tool inventory
- ✅ Organizes toolbox logically
- ✅ Tools accessible and visible
- ✅ Heavy items safely stored (bottom)
- ✅ Can locate any tool quickly

Hour 2 (2:00 PM - 3:00 PM): Tool Maintenance Practice

Activity 1: Hand Tool Inspection & Maintenance (25 min)

- Students inspect tools for common problems:

Wrench inspection:

- [ ] Check for cracks (hold up to light, look at stressed areas)
- [ ] Check jaws (not rounded or damaged)
- [ ] Check for bending (wrench should be straight)
- [ ] Clean and wipe with oil-soaked rag
- [ ] **Action:** Remove any cracked wrenches from service

Socket inspection:

- [ ] Check corners (not rounded - 12-point especially vulnerable)
- [ ] Check for cracks around drive hole
- [ ] Clean inside and out
- [ ] Light oil if storing long-term

- [ ] **Action:** Replace rounded sockets

Ratchet maintenance:

- [ ] Check operation (smooth, consistent clicks)
- [ ] Reverse lever works smoothly
- [ ] Clean exterior
- [ ] Add few drops of oil to mechanism
- [ ] Cycle several times to distribute oil
- [ ] Wipe excess

Screwdriver inspection:

- [ ] Check tip (not rounded, chipped, or broken)
- [ ] Check handle (not cracked)
- [ ] Clean shaft
- [ ] Lightly oil if storing
- [ ] **Action:** Grind/file tip if slightly damaged, replace if severe

Activity 2: Measuring Tool Maintenance (20 min)

- **Caliper maintenance practice:**
  1. Clean jaws with alcohol (remove any oil, debris)
  2. Check for damage (bent jaws, damaged scale)
  3. Check zero:
     - Close jaws fully
     - Should read 0.00 mm
     - If not, may have zero-set button, or needs service
  4. Open and close smoothly (no binding)
  5. Apply light oil to sliding surface
  6. Close jaws when storing
  7. Store in case
- **Micrometer maintenance practice:**
  1. Clean anvil and spindle faces (alcohol, soft cloth)
  2. Check calibration:
     - Close spindle to anvil (zero position)
     - Should read 0.000 mm
     - Or use calibration gauge block
  3. Check smooth operation (spindle rotates freely)
  4. Light oil on threads
  5. Store in case
  6. Never force closed

Activity 3: Torque Wrench Maintenance (15 min)

- **Critical maintenance procedure:**

Proper torque wrench care:

1. **After use:**
   - Wipe clean (remove oil, grease)
   - Return to LOWEST setting
     - Rotate handle fully counterclockwise
     - Relieves spring tension
     - Prevents calibration drift
2. **Storage:**
   - Store horizontally in case
   - Don't hang (affects calibration)
   - Protect from moisture
3. **Calibration check:**
   - Instructor demonstrates calibration verification:
     - Use calibration tester or beam-type wrench
     - Set to 100 Nm
     - Apply to calibration device
     - Should click at 100 Nm ±4% (96-104 Nm acceptable)
     - If outside range, needs professional calibration

Students practice:

- Return torque wrench to lowest setting
- Store properly in case
- Understand why this is critical
- Recognize calibration as professional service (not DIY)

Pass Criteria:

- ✅ Can inspect tools for common damage
- ✅ Identifies tools needing replacement
- ✅ Cleans and maintains tools properly
- ✅ Checks caliper zero correctly
- ✅ Returns torque wrench to lowest setting
- ✅ Stores measuring tools in cases

Hour 3 (3:00 PM - 4:00 PM): Parts Catalog & Service Information

Activity 1: VIN Decoding Practice (20 min)

- **Students practice finding and decoding VINs**

Exercise 1: Locate VIN on vehicle

- Find VIN in 3 locations:
  1. Dashboard (through windshield)
  2. Driver door jamb (sticker)
  3. Registration papers
- Verify all match
- Photograph locations

Exercise 2: Decode VIN

- Given VIN: MA3ERLF1S00289764
- Students identify:
  - Manufacturer: MA3 = Maruti Suzuki
  - Model/variant: ERLF1 = (decode using chart)
  - Year: S = 2018 (or 1995 - verify with other info)
  - Plant: 0 = Specific plant
  - Serial number: 289764

Exercise 3: Use VIN for information

- Enter VIN in online decoder (free websites available)
- System provides:
  - Make, Model, Year
  - Engine type and displacement
  - Transmission type
  - Trim level
  - Factory options
- Verify matches actual vehicle

Activity 2: Parts Catalog Navigation (25 min)

- **Using online parts catalog (demo system or actual)**

Exercise: Find brake pads for specific vehicle

1. **Enter vehicle details:**
   - VIN: [actual vehicle VIN]
   - Or select: Make > Model > Year > Engine
2. **Navigate to brake system:**

- Categories > Chassis > Brake System > Front Brakes

3. **View diagram:**
   - Exploded view of brake assembly
   - Parts numbered
4. **Select brake pads:**
   - Click on pad in diagram
   - Or search list
5. **Get part information:**
   - Part number: Example: 04465-0K260
   - Description: Front Brake Pad Set
   - Price: ₹2,850
   - Availability: In stock
   - Fitment notes: Verify vehicle has 16" wheels
6. **Cross-reference:**
   - OEM part: 04465-0K260
   - Aftermarket alternatives:
     - Bendix DB1234
     - Bosch 0986AB2345
     - Compare quality and price

Students practice finding parts:

- Oil filter for their practice vehicle
- Air filter
- Spark plugs
- Wiper blades
- Each student finds different parts, shares part numbers

Activity 3: Service Information Access (15 min)

- **Accessing repair procedures online**

Exercise: Find oil change procedure

1. Access service information system (demo or actual)
2. Select vehicle
3. Navigate: Maintenance > Scheduled Service > Engine Oil Change
4. Review procedure:
   - Required tools
   - Parts needed
   - Step-by-step instructions
   - Torque specifications

- Capacities
- Warnings/cautions

Students locate information:

- Coolant capacity for specific vehicle
- Spark plug torque specification
- Brake pad minimum thickness
- Tire pressure specification
- Engine oil viscosity grade

Compare sources:

- Owner's manual (basic info)
- Service manual (detailed procedures)
- Parts catalog (part numbers, diagrams)
- **Each has different purpose, use appropriate source**

Pass Criteria:

- ✅ Can locate VIN on vehicle
- ✅ Can decode basic VIN information
- ✅ Can navigate parts catalog system
- ✅ Can find correct parts using VIN
- ✅ Can access service information online
- ✅ Can locate specifications and procedures

DAY 27 ASSESSMENT:

Practical Skills Checklist:

- [ ] Creates complete tool inventory
- [ ] Organizes toolbox logically and efficiently
- [ ] Inspects tools for damage
- [ ] Performs proper tool maintenance
- [ ] Cleans and stores measuring tools correctly
- [ ] Returns torque wrench to lowest setting
- [ ] Locates VIN on vehicle (multiple locations)
- [ ] Decodes VIN information
- [ ] Navigates parts catalog software
- [ ] Finds correct parts using VIN

- [ ] Accesses online service information
- [ ] Locates specifications and procedures

Knowledge Check (Verbal):

- Why should torque wrench be returned to lowest setting?
- How often should torque wrenches be calibrated?
- What does VIN stand for and what does it tell you?
- Why is using VIN better than using make/model/year for parts?
- Name three sources of service information

DAY 28: CUSTOMER COMMUNICATION + SERVICE WRITING

Learning Objectives:

- Practice professional customer interaction
- Learn to explain technical issues in simple terms
- Write clear service recommendations
- Create accurate estimates
- Handle customer concerns professionally
- Build trust and repeat business

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Customer Communication Fundamentals

Why Communication Skills Matter:

- **Technical skill ≠ Communication skill**
- Great technician + poor communication = customer dissatisfaction
- Average technician + great communication = customer loyalty
- **Communication builds trust**
- Trust = repeat business = career success

First Impressions:

- **Appearance matters:**
  - Clean uniform
  - Name tag visible

    - Neat, professional appearance
    - Workshop clean and organized
- **Greeting:**
    - Smile, make eye contact
    - Introduce yourself: "Good morning, I'm Raj, I'll be servicing your vehicle today"
    - Positive attitude
    - Confident but friendly

Active Listening:

- **Let customer explain completely** (don't interrupt)
- **Take notes** (shows you care, ensures accuracy)
- **Ask clarifying questions:**
    - "When does the noise occur?"
    - "How long has this been happening?"
    - "Can you describe the sound?"
- **Repeat back:** "So you're hearing a squealing noise when you brake, especially in the morning?"
- **Verify understanding** before customer leaves

Customer Concerns (The 3-C's):

1. **Concern:** What customer noticed/experienced
2. **Cause:** What technician diagnoses
3. **Correction:** What needs to be done

Professional Language:

- **Use simple terms, not jargon**
- ❌ "Your O2 sensor is giving a P0420 code"
- ✅ "Your emission system has a faulty sensor that needs replacement"

Translation Examples:

- **Technical:** "Brake pads are at 2mm thickness, below minimum specification of 3mm"
- **Customer:** "Your brake pads are worn out and need to be replaced for safe braking"
- **Technical:** "Front lower control arm bushings are deteriorated, causing excessive play"
- **Customer:** "The rubber parts that connect your suspension to the frame are worn out, causing loose handling"
- **Technical:** "Coolant shows high conductivity indicating electrolysis, thermostat stuck open showing 72°C operating temperature"
- **Customer:** "Your cooling system fluid is contaminated and the temperature control valve isn't working properly"

Show, Don't Just Tell:

- **Visual demonstration powerful:**
  - Show worn brake pad next to new one
  - Show customer the leak
  - Show photos taken during inspection
  - Point to worn tire tread
- **Helps customer understand:**
  - Why repair needed
  - Urgency level
  - Value of service
- **Increases approval rate** significantly

Building Trust:

- **Honesty first:**
  - Never recommend unnecessary work
  - Admit if you don't know (then find out)
  - Explain options (OEM vs aftermarket)
  - Give realistic timelines
- **Transparency:**
  - Explain what you're doing and why
  - Show findings before starting work
  - Call if additional work needed
  - No surprises on bill
- **Follow through:**
  - If you say 2 hours, deliver in 2 hours
  - If you promise to call, call
  - Quality work = trust

Hour 2 (10:00 AM - 11:00 AM): Writing Service Recommendations

Components of Good Recommendation:

1. **What's wrong** (finding)
2. **Why it matters** (consequence)
3. **What to do** (solution)
4. **When to do it** (urgency)
5. **How much** (cost)

Example - Good Recommendation:

```
FINDING: Front brake pads measured at 2mm thickness

WHY IT MATTERS: Minimum safe thickness is 3mm. At 2mm, the metal wear
indicator is touching the rotor, causing the squeaking noise. If not
replaced soon, the pads will wear through completely, damaging the
rotors and creating a safety hazard.

SOLUTION: Replace front brake pads with genuine parts

URGENCY: Recommend within this week (URGENT - Safety)

COST: Parts ₹2,500 + Labor ₹800 = ₹3,300 + taxes
```

Example - Poor Recommendation: "Need front brakes"

- Vague, no explanation, no context, no urgency, no cost

Prioritization System:

🔴 **RED (Urgent - Do Now):**

- Safety critical
- Illegal to drive
- Will cause breakdown soon
- Will damage other parts if delayed
- **Examples:**
  - Brake pads 1mm (unsafe)
  - Tire tread 1.5mm (illegal)
  - Major oil leak (engine damage imminent)
  - Check engine light flashing (catalyst damage occurring)

🟡 **YELLOW (Recommended - Do Soon):**

- Will worsen if ignored
- Not immediate safety issue
- Schedule within 1-4 weeks
- Prevent more expensive repairs
- **Examples:**
  - Brake pads 3mm (will need soon)
  - Small oil leak (monitor, will worsen)
  - Worn suspension bushings (affects handling)
  - Battery 4 years old, low voltage (may fail)

🟢 GREEN (Monitor - Check Next Service):

- Working but showing wear
- Adequate life remaining
- Check again at next service
- Plan for future expense
- **Examples:**
  - Brake pads 5mm (good for now)
  - Tires 4mm tread (good for 10,000 km)
  - Battery 2 years old, testing good
  - Coolant slightly dirty but adequate

Creating Estimates:

Detailed estimate format:

```
ESTIMATE - Front Brake Service

PARTS:
- Brake pad set (front): ₹2,500
- Brake cleaner: ₹150
- Anti-squeal compound: ₹200
Subtotal Parts: ₹2,850

LABOR:
- Remove wheels: 0.2 hrs
- Replace brake pads: 0.8 hrs
- Inspect/clean calipers: 0.3 hrs
- Test drive: 0.2 hrs
Total Labor: 1.5 hrs @ ₹800/hr = ₹1,200

SUBTOTAL: ₹4,050
GST (18%): ₹729
TOTAL ESTIMATE: ₹4,779

Note: Estimate valid for 30 days. Additional work may be required
if rotors are found to be below minimum thickness during service.
```

Estimate Best Practices:

- **Be specific** (not "brake job" - list actual parts)
- **Include labor time** (shows you're not guessing)
- **Note limitations** ("estimate pending inspection")

- **Get approval** before exceeding estimate
- **Provide written copy** to customer

Upselling vs Recommendations:

- **Upselling = selling unnecessary services** ❌
- **Recommendations = informing about needed services** ✅
- **Big difference:**
  - Upsell: Customer doesn't need it, you push it
  - Recommend: Customer needs it, you inform them
- **Always ethical:** Only recommend what's actually needed

The "Now vs Later" Conversation:

```
Customer: "Do I really need to do this today?"

Good response: "The front brakes are a safety issue - I strongly
recommend doing those today. The rear brakes have some life left,
so we can monitor those and do them in a few months when they're
closer to the minimum. Would you like to do the fronts today and
schedule the rears for your next service?"

Bad response: "Yes, you have to do everything I said."
```

Give customers choices when appropriate:

- OEM vs quality aftermarket parts
- Now vs schedule for next week
- Package deals (do all four tires vs just two)
- Payment plans (if available)

Hour 3 (11:00 AM - 12:00 PM): Handling Difficult Situations

Common Customer Objections:

1. "That's too expensive!"

- **Bad response:** "That's the price, take it or leave it"
- **Good response:**
  - Explain value: "These are genuine parts with 2-year warranty..."
  - Break down cost: "The parts are ₹X, labor is ₹Y..."

  - Offer options: "We could use aftermarket parts for ₹X less, but they only have 1-year warranty..."
  - Explain consequence of delay: "I understand it's an investment. However, if we don't replace the brake pads now, you'll damage the rotors, which will increase the cost to ₹X..."

2. "My friend/YouTube says I don't need that"

- **Bad response:** "Your friend doesn't know what they're talking about"
- **Good response:**
  - Show evidence: "Let me show you the actual brake pad - see how thin it is compared to this new one?"
  - Explain professionally: "I can show you the measurement - it's at 2mm, and the minimum safe thickness is 3mm"
  - Stay confident but respectful: "I understand there's a lot of information out there. I'm showing you what I found on your specific vehicle today"

3. "I just want what I asked for, don't try to sell me more"

- **Bad response:** *Push harder*
- **Good response:**
  - Respect request: "I understand. I'll do exactly what you've asked for"
  - Fulfill responsibility: "However, as part of my professional responsibility, I need to inform you that I found [issue]. Here's a written report of what I found. The decision is completely yours"
  - Document: Make sure customer signs acknowledgment that they declined recommendations

4. "The last shop didn't say I needed this"

- **Bad response:** "They didn't know what they were doing"
- **Good response:**
  - Acknowledge: "That may be true - sometimes things wear quickly, or they may have missed it"
  - Show current condition: "What I can show you is the current condition today..."
  - Stay professional: Don't criticize other shops

5. "Why didn't you call me before doing this?"

- **This is YOUR fault - you should have called**
- **Response:**
  - Apologize: "You're absolutely right, I should have called before proceeding. I apologize"

  - Explain: "I found [issue] and thought it was included in the original authorization, but I should have confirmed"
  - Make it right: "I take full responsibility. Let's discuss how to resolve this"

When Customer is Angry:

- **Don't take it personally** (they're angry at situation, not you)
- **Stay calm** (don't get defensive or angry back)
- **Listen first** (let them vent)
- **Empathize:** "I understand you're frustrated, I would be too"
- **Take responsibility** if shop at fault
- **Offer solution:** "Here's what I can do to make this right..."
- **Get supervisor** if needed (not a sign of weakness)

When You Made a Mistake:

- **Own it immediately** (don't hide or blame others)
- **Apologize sincerely**
- **Explain what happened** (briefly, not making excuses)
- **Fix it at no charge**
- **Learn from it** (how to prevent in future)
- **Follow up** (ensure customer satisfied)
- **Honesty builds long-term trust** even when you mess up

Phone Etiquette:

- **Answer professionally:** "Good morning, [Shop Name], this is Raj speaking, how may I help you?"
- **Take messages accurately:** Write down name, number, message, time
- **Return calls promptly** (within 2 hours if possible)
- **Call with updates:** "Just wanted to let you know we found the problem..."
- **Bad news delivered personally** (don't text "Your engine is blown")

Written Communication:

- **Professional email/text:**
  - Proper grammar and spelling
  - Clear and concise
  - Include relevant details (vehicle, service)
  - Professional closing
- **Example:**

```
Dear Mr. Sharma,

Your 2019 Honda City service is complete. We performed the 60,000 km
service including oil change, filter replacements, and multi-point
inspection.

During inspection, we found the front brake pads are at 3mm thickness.
They are still safe for now but will need replacement within the next
10,000 km. I've attached photos for your reference.

Your vehicle is ready for pickup. Total: ₹5,240 (as estimated).

Please call if you have any questions.

Best regards,
Raj Kumar
ABC Auto Service
Ph: 9876543210
```

Building Long-Term Relationships:

- **Remember regular customers** (names, vehicles, preferences)
- **Service reminders:** Text/email when service due
- **Follow up:** "How's the car running after the service?"
- **Honesty over short-term profit**
- **Consistency:** Every service is quality
- **Small touches:** Clean car, adjusted mirrors, service sticker
- **Customers become advocates** (referrals)

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Customer Interaction Roleplay

Activity 1: Initial Customer Greeting (20 min)

- **Students practice professional greetings**

Scenario: Customer arrives for oil change appointment

Student (technician) must:

1. Greet customer professionally

2. Introduce themselves
3. Confirm appointment
4. Ask about concerns
5. Explain what will be done
6. Set expectations (time)
7. Get contact number

Example:

```
Technician: "Good morning! Welcome to ABC Auto. I'm Rajesh, I'll be
taking care of your vehicle today. I see you're scheduled for an oil
change at 10:00 AM?"

Customer: "Yes, that's right."

Technician: "Perfect. We'll do the oil change and our complete multi-point
inspection. It should take about 45 minutes. Is there anything specific
you'd like us to check while we have it?"

Customer: "Well, there's a squeak when I brake sometimes."

Technician: "Okay, I'll make a note of that and inspect the brakes
carefully. Let me get your contact number in case we find anything
that needs your approval."

[Takes information]

Technician: "Great. You're welcome to wait in our customer lounge, or
we can call you when it's ready. We'll also call if we find anything
unexpected that needs your attention."
```

Students practice in pairs:

- Rotate roles (technician/customer)
- Instructor observes and provides feedback
- Focus on eye contact, tone, confidence

Activity 2: Explaining Findings Roleplay (40 min)

- **Practice translating technical findings to customer language**

Scenario 1: Worn Brake Pads

```
Technical finding: Front brake pads 2mm, minimum 3mm, wear indicator
contacting rotor causing audible squeal

Student must explain to "customer" in simple terms:
- What was found
- Why it's a problem
- What needs to be done
- Cost
- Urgency
```

Good example: "Mr. Sharma, I found the source of that squeaking noise. Your front brake pads are worn down to 2 millimeters thick - for safety they should be at least 3mm. Here, let me show you [shows photo or actual part]. See how thin this is compared to a new pad? That metal piece you see is the wear indicator

- it's designed to squeal when the pad gets too thin, warning you it's time to replace them. I recommend replacing them this week - they're still safe for a few hundred kilometers, but if we wait too long, you'll damage the metal rotors which are expensive to replace. The pads will cost ₹2,500 for parts plus ₹800 labor. Would you like me to take care of this today?"

Scenarios for practice:

1. Worn brake pads (above)
2. Low tire tread with uneven wear (needs alignment too)
3. Old battery failing voltage test
4. Oil leak from valve cover gasket
5. Worn suspension bushings

Each student practices:

- Instructor or another student plays customer
- Must use simple language
- Must show evidence (photo, parts, measurements)
- Must explain urgency correctly
- Must give accurate cost estimate

Instructor evaluates:

- ✅ Clear, simple language
- ✅ Shows evidence
- ✅ Explains consequence
- ✅ Gives options if appropriate

- ✅ Professional, confident delivery

Pass Criteria:

- ✅ Greets customer professionally
- ✅ Listens actively and takes notes
- ✅ Explains findings in simple, clear language
- ✅ Shows evidence to support findings
- ✅ Prioritizes correctly (urgent/soon/monitor)
- ✅ Provides accurate cost estimates
- ✅ Maintains professional demeanor

Hour 2 (2:00 PM - 3:00 PM): Service Writing Practice

Activity 1: Writing Customer-Facing Recommendations (30 min)

- **Students convert technical findings to written recommendations**

Given scenario:

```
Vehicle: 2018 Maruti Swift, 65,000 km
Customer concern: "Check engine light is on"
Technical findings:
- P0420 - Catalyst efficiency below threshold
- Oxygen sensor voltage readings:
  - Upstream cycling normally 0.1-0.9V
  - Downstream stuck at 0.45V (should be stable ~0.45V but reading
identical to upstream)
- Visual inspection: Catalyst intact, no external damage
- No exhaust leaks found
- Engine running smoothly, no misfires
```

Students write recommendation:

Poor example: "Replace downstream O2 sensor, code P0420"

Good example:

```
FINDING: Check Engine Light - Code P0420
Your vehicle's emission system monitoring has detected that the
catalytic converter (the component that cleans exhaust gases) is not
working as efficiently as it should.
```

```
Our testing shows the sensor after the catalytic converter is giving
incorrect readings. This sensor is supposed to verify the catalyst is
working, but it's providing faulty information.

RECOMMENDATION: Replace downstream oxygen sensor
This will restore proper monitoring and should turn off the check
engine light. The catalytic converter itself appears to be functioning
correctly based on our tests.

URGENCY: RECOMMENDED (Soon - within 2 weeks)
While not an immediate safety concern, you will fail emission testing
with this code, and it prevents the system from detecting other
potential emission problems.

COST ESTIMATE:
- Oxygen sensor: ₹3,500
- Labor (1.0 hr): ₹800
Subtotal: ₹4,300
Tax (18%): ₹774
Total: ₹5,074

If the light returns after sensor replacement, the catalytic converter
itself may need replacement (₹18,000-25,000), but we'll verify this
first with the less expensive sensor replacement.
```

Students practice writing for:

- Suspension noise (worn shock absorbers)
- AC not cooling (refrigerant leak)
- Hard starting when cold (coolant temp sensor)

Peer review:

- Exchange written recommendations
- Check for:
  - Clear explanation
  - Appropriate detail (not too much, not too little)
  - Correct prioritization
  - Accurate cost estimate
  - Professional language

Activity 2: Creating Multi-Item Service Recommendations (30 min)

- **Complex scenario with multiple findings**

Scenario:

```
Vehicle: 2017 Honda City, 78,000 km, 5 years old
Findings:
1. Front brake pads: 3mm (minimum 3mm)
2. Rear brake pads: 6mm
3. All tires: 4mm tread, uneven front wear (inside edge)
4. Battery: 5 years old, 11.9V (engine off), load test marginal
5. Coolant: Never changed, dark brown color
6. Small oil seep at valve cover gasket (few drops on ground)
7. Cabin air filter: Heavily contaminated
```

Students create prioritized recommendation document:

```
SERVICE RECOMMENDATIONS
2017 Honda City | 78,000 km | VIN: XXXXX

🔴 URGENT (Recommend Today):
1. Battery Replacement
   Current: 11.9V (low), 5 years old, failing load test
   Why: May fail to start, especially in cold weather
   Cost: Battery ₹4,500 + Installation ₹300 = ₹4,800

🟡 RECOMMENDED (Within 1 Month):
2. Front Brake Pads
   Current: 3mm (at minimum safe thickness)
   Why: Safety critical, will damage rotors if delayed
   Cost: Pads ₹2,500 + Labor ₹800 = ₹3,300

3. Coolant System Service
   Current: Original coolant, 5 years old, dark/contaminated
   Why: Corrosion protection depleted, can damage engine
   Cost: Coolant ₹800 + Labor ₹1,200 = ₹2,000

4. Wheel Alignment
   Current: Uneven front tire wear indicates misalignment
   Why: Extends tire life, improves handling
   Cost: 4-wheel alignment ₹1,500

5. Cabin Air Filter Replacement
   Current: Heavily contaminated
```

```
   Why: Poor airflow, musty smell, allergens
   Cost: Filter ₹450 + Labor ₹200 = ₹650

🟢 MONITOR (Check Next Service):
6. Rear Brake Pads
   Current: 6mm (adequate life remaining ~20,000 km)
   When: Check at 90,000 km service

7. Valve Cover Gasket (Oil Seep)
   Current: Minor seepage, not dripping actively
   When: Monitor, may need replacement in 6-12 months (₹2,500)

8. Tires
   Current: 4mm tread (adequate for ~15,000 km)
   When: Plan replacement around 90,000 km (₹20,000 for set of 4)

SUMMARY:
Urgent work today: ₹4,800
Recommended within month: ₹7,450
Total if all done: ₹12,250 + tax

Would you like to proceed with the urgent battery replacement today,
and schedule the other items for next week?
```

Pass Criteria:

- ✅ Writes clear, customer-friendly recommendations
- ✅ Includes all necessary information (finding, why, cost, urgency)
- ✅ Prioritizes correctly using 3-tier system
- ✅ Provides accurate cost estimates
- ✅ Offers options when appropriate
- ✅ Professional formatting and language

Hour 3 (3:00 PM - 4:00 PM): Handling Objections Roleplay

Activity 1: Common Objection Scenarios (45 min)

- **Students practice handling difficult conversations**

Scenario 1: "That's too expensive!"

- Customer shocked by brake replacement cost (₹8,000 for all four corners)

- Student must:
  - Show empathy
  - Break down cost
  - Explain value
  - Offer options
  - Explain consequences of delay

Scenario 2: "I'll get a second opinion"

- Customer doubts diagnosis of bad alternator
- Student must:
  - Stay professional (not defensive)
  - Offer to show test results
  - Respect customer's decision
  - Provide written documentation
  - Invite customer back

Scenario 3: "Why didn't this last longer?"

- Customer angry that battery failed after only 3 years
- Student must:
  - Empathize
  - Explain typical life (3-5 years normal)
  - Discuss warranty if applicable
  - Not promise unrealistic longevity
  - Focus on solution, not blame

Scenario 4: "I only came for oil change, stop trying to sell me things"

- Customer upset about additional recommendations
- Student must:
  - Respect boundary
  - Explain professional responsibility to inform
  - Provide written report
  - Offer to show findings
  - Don't pressure

Scenario 5: "The internet says this should only cost ₹X"

- Customer comparing prices from online sources
- Student must:
  - Acknowledge price awareness
  - Explain quality differences (OEM vs cheap parts)

- Explain warranty included
- Explain labor rate justification (skilled technician, proper tools)
- Stand by pricing confidently

Each scenario:

- Student plays technician
- Instructor or another student plays difficult customer
- 5 minutes per scenario
- Class observes and provides feedback
- Discuss what worked, what didn't

Activity 2: Creating Phone Scripts (15 min)

- **Students write scripts for common phone calls**

Script 1: Calling customer with additional findings

```
"Hello Mr. Sharma, this is Rajesh from ABC Auto calling about your Honda
City. I wanted to update you on the service.

The oil change is complete, but during our inspection I found that the
front brake pads are very worn - they're at 2mm which is below the safe
minimum of 3mm. This is causing the squeaking noise you mentioned.

I'd like to show you photos and the actual measurement. The pads need to
be replaced for safe braking. The cost would be ₹3,300 for parts and
labor.

I haven't started any additional work - I wanted to get your approval
first. When would be a good time for you to stop by so I can show you
what I found? Or if you prefer, I can send you the photos and we can
discuss over the phone."
```

Script 2: Calling with delay notification

```
"Hello Mrs. Patel, this is Rajesh from ABC Auto. I wanted to call and
let you know that your vehicle will take about 30 minutes longer than
we originally estimated.

When we started the brake work, we discovered the caliper slide pins were
seized, which is why you were experiencing the pulling sensation. We need
to service the calipers to make sure the brakes work properly and evenly.
```

```
This is included in the brake service - no additional charge. I just
wanted
to keep you informed. Your vehicle should be ready around 2:30 PM instead
of 2:00 PM. Is that okay with your schedule?"
```

Students practice reading scripts:

- Professional tone
- Clear delivery
- Empathy and confidence
- Take turns, provide feedback

Pass Criteria:

- ✅ Handles objections professionally
- ✅ Stays calm under pressure
- ✅ Doesn't take criticism personally
- ✅ Provides factual responses
- ✅ Offers solutions, not arguments
- ✅ Knows when to get supervisor help
- ✅ Communicates clearly on phone
- ✅ Updates customers proactively

DAY 28 ASSESSMENT:

Practical Skills Checklist:

- [ ] Greets customers professionally
- [ ] Listens actively and takes accurate notes
- [ ] Explains technical findings in simple language
- [ ] Shows evidence to support recommendations
- [ ] Prioritizes recommendations correctly (urgent/soon/monitor)
- [ ] Provides accurate cost estimates
- [ ] Writes clear, customer-friendly service recommendations
- [ ] Handles customer objections professionally
- [ ] Maintains composure with difficult customers
- [ ] Communicates effectively on phone
- [ ] Creates professional written documentation

Knowledge Check (Verbal):

- What are the 3 C's of customer service?
- How do you translate technical jargon for customers?
- What's the difference between upselling and making recommendations?
- How should you handle an angry customer?
- Why is it important to show evidence rather than just tell?

DAY 29: CROSS-SYSTEM INTEGRATION + COMPLEX SCENARIOS

Learning Objectives:

- Apply knowledge from all four technical weeks together
- Diagnose problems affecting multiple systems
- Practice systematic troubleshooting methodology
- Execute complete vehicle assessments
- Integrate professional practices with technical skills

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Systematic Troubleshooting Methodology

The Problem-Solving Framework:

Why Systematic Approach Matters:

- **Random testing wastes time and money**
- **Systematic approach finds root cause efficiently**
- **Prevents misdiagnosis and comebacks**
- **Professional vs amateur difference**
- **Builds customer confidence**

The 8-Step Diagnostic Process:

Step 1: Verify the Complaint

- **Never assume customer description is accurate**
- Experience the problem yourself when possible
- Ask clarifying questions:
  - When does it occur? (cold start, hot engine, all the time)
  - Where does it occur? (specific speed, turning, braking)

    - What exactly happens? (noise, vibration, performance issue)
    - How long has it been happening?
    - Any recent work done?
- Test drive if symptom is operational
- Document exactly what you observe

Step 2: Gather Information

- **Vehicle history:**
    - Check service records
    - Previous repairs/complaints
    - Maintenance status
- **Technical resources:**
    - Service bulletins (TSBs)
    - Common problems for this model
    - Recall information
    - Online forums/databases
- **Related systems:**
    - What systems could cause this symptom?
    - List all possibilities

Step 3: Research Possible Causes

- Create list of potential causes
- Consider:
    - Most common failures first (horses, not zebras)
    - Recent work (did someone touch this system?)
    - Related symptoms
    - System interdependencies
- Narrow down based on:
    - When problem occurs
    - Conditions that trigger it
    - What makes it better/worse

Step 4: Isolate the Problem

- **Test most likely causes first**
- Use proper diagnostic procedures:
    - Visual inspection
    - Basic tests (voltage, pressure, etc.)
    - Scan tool data
    - Service manual procedures

- Eliminate possibilities systematically
- Don't skip steps (thoroughness over speed)

Step 5: Identify Root Cause

- **Symptom vs cause:**
  - Low coolant = symptom
  - Leaking water pump = cause
- **Don't treat symptoms:**
  - Adding coolant doesn't fix leak
  - Must repair water pump
- Verify root cause:
  - Test confirms diagnosis
  - Makes logical sense
  - Explains all symptoms

Step 6: Develop Repair Plan

- What needs to be fixed/replaced
- Parts required (part numbers, availability)
- Special tools needed
- Labor time estimate
- Sequence of operations
- Additional items to check while apart
- Total cost estimate

Step 7: Repair and Verify

- Perform repair properly (torque specs, procedures)
- Quality check work
- Test operation
- Verify problem resolved
- Check for related issues
- Test drive (if applicable)

Step 8: Document and Follow Up

- Complete service order:
  - Customer concern
  - Cause found
  - Correction performed
- Document tests performed
- Note any recommendations

- Follow up with customer
- Ensure satisfaction

Hour 2 (10:00 AM - 11:00 AM): Understanding System Interactions

How Vehicle Systems Interconnect:

Case Study 1: Engine Performance Problem

```
Symptom: Poor acceleration, black smoke, reduced fuel economy

Possible Systems Involved:
1. AIR INTAKE SYSTEM
   - Clogged air filter → not enough air → rich mixture → black smoke
   - MAF sensor dirty → incorrect air measurement → wrong fuel amount
   - Intake leak → unmetered air → lean mixture BUT O2 compensates to rich

2. FUEL SYSTEM
   - High fuel pressure → too much fuel → rich mixture
   - Leaking injectors → excess fuel
   - Fuel pressure regulator failed

3. EXHAUST/EMISSION SYSTEM
   - Clogged catalyst → back pressure → reduced power
   - Failed O2 sensor → ECU can't adjust mixture properly

4. IGNITION SYSTEM
   - Weak spark → incomplete combustion → unburned fuel → black smoke
   - Wrong spark plugs → poor combustion

5. ENGINE MECHANICAL
   - Low compression → poor combustion → black smoke
   - Worn piston rings → oil burning (blue smoke) or poor compression

Systematic Approach:
- Start simple: Check air filter (5 minutes, free)
- Scan for codes: Check O2 sensor data, fuel trim
- Test fuel pressure: Within spec?
- Inspect spark plugs: Condition tells story
- Compression test: If nothing else found

Root Cause Example: Dirty MAF sensor
```

```
- Measures less air than actually entering
- ECU adds excess fuel to match "low" air reading
- Results in rich mixture, black smoke, poor economy
- Fix: Clean MAF sensor (10 minutes, ₹100)
```

Case Study 2: Overheating Problem

```
Symptom: Engine temperature gauge in red zone

Multiple Systems Can Cause This:

1. COOLING SYSTEM (Primary suspect)
   - Low coolant level → insufficient heat transfer
   - Failed thermostat (stuck closed) → no radiator flow
   - Clogged radiator → reduced heat dissipation
   - Failed water pump → no circulation
   - Radiator fan not working → no airflow at idle
   - Air in system → prevents proper circulation
   - Collapsed hose → flow restriction

2. IGNITION SYSTEM
   - Retarded timing → combustion too late → excess heat
   - Can cause overheating even with good cooling system

3. EXHAUST SYSTEM
   - Clogged catalyst → back pressure → engine works harder → more heat

4. ENGINE MECHANICAL
   - Head gasket leak → combustion gases in coolant → overheats rapidly
   - Can also cause coolant in oil (milky appearance)

5. LUBRICATION SYSTEM
   - Low oil → friction → heat (but shows as oil pressure problem usually)

Diagnosis Sequence:
1. Verify coolant level (engine cold)
2. Pressure test cooling system (finds leaks)
3. Check fan operation (should run when hot)
4. Test thermostat (should open at spec temp)
5. Check for exhaust gases in coolant (head gasket test)
6. Inspect water pump (look for leaks, listen for bearing noise)

Root Cause Example: Failed thermostat (stuck closed)
```

```
- Coolant can't reach radiator
- All heat stays in engine block
- Overheats quickly even though radiator cold
- Fix: Replace thermostat (₹500 part + 1 hr labor)
```

Case Study 3: No-Start Condition

```
Symptom: Engine cranks but won't start

Multiple Systems Required for Starting:

1. FUEL SYSTEM
   - Must have: Fuel in tank, fuel pressure, injector pulse
   - Check: Fuel pump runs? Pressure at rail? Injectors clicking?

2. IGNITION SYSTEM
   - Must have: Spark at plugs, correct timing
   - Check: Spark present? Coils firing? CKP sensor signal?

3. AIR INTAKE
   - Must have: Air entering engine
   - Usually not the problem, but check for severe restriction

4. ENGINE MECHANICAL
   - Must have: Compression in cylinders
   - Check if: Timing belt broke (interference engine = bent valves)

5. ELECTRICAL/COMPUTER
   - Must have: Good battery, communication, immobilizer off
   - Check: Battery voltage adequate? Security light? Codes stored?

6. EMISSION SYSTEM
   - Rarely prevents starting, but sensor failures can
   - Check: CKP sensor (engine needs this), CMP sensor (some vehicles)

Three Things Needed to Start:
1. AIR (intake open)
2. FUEL (pressure and injection)
3. SPARK (at right time)

Missing ANY of these three = no start

Diagnosis Strategy:
```

```
- Check easiest first:
  - Gas in tank? (customer error common)
  - Battery charged? (12.4V minimum)
  - Security light on? (immobilizer active)
- Then systematic:
  - Fuel pressure test
  - Spark test
  - Compression test (if needed)
- Scan for codes (helps narrow down)

Root Cause Example: Failed crankshaft position sensor
- ECU doesn't know engine position/speed
- Can't time fuel injection or spark
- Engine cranks but no start
- Usually sets code P0335 or P0340
- Fix: Replace CKP sensor (₹1,500-3,000)
```

Hour 3 (11:00 AM - 12:00 PM): Complex Diagnostic Scenarios

Intermittent Problems - Most Difficult:

Why Intermittent Problems Are Hard:

- Doesn't happen when testing
- Can't verify repair (was it really fixed?)
- Customer frustration (shop "can't find problem")
- Multiple visits often required

Strategies for Intermittent Issues:

1. Detailed History:

- **Exactly** when does it occur?
  - Time of day? (morning cold start, afternoon heat)
  - Weather related? (rain, humidity, temperature)
  - After driving how long?
  - Specific conditions? (bumps, turns, braking)
- Pattern recognition crucial

2. Data Logging:

- Some scan tools can record data

- Customer drives with scanner
- Captures conditions when problem occurs
- Review data to identify cause

3. Component Testing at Limits:

- Heat gun to warm sensors (does problem occur?)
- Cooling spray to cool components
- Wiggle test on wiring (intermittent connection?)
- Road test over bumps (loose connection?)

4. Statistical Approach:

- Replace most common failure items
- Monitor to see if problem returns
- Not ideal, but sometimes necessary

Example: Intermittent Stalling

```
Customer complaint: "Car sometimes stalls when stopping at lights,
but only when it's hot outside, and only after driving 30+ minutes.
Doesn't happen every day."

Challenge: Can't reproduce in shop (too time consuming)

Investigation approach:
1. Check history: Any pattern to failures?
2. Scan for codes: May have pending codes even if no check light
3. Review freeze frame: Conditions when last occurred?
4. Check recalls/bulletins: Known issue for this model?
5. Inspect common failures:
   - Idle air control valve (carbon buildup)
   - Throttle position sensor (worn spot)
   - Crankshaft position sensor (heat sensitive)
   - Fuel pump (weak when hot)
6. Test drive: Try to replicate conditions
7. Monitor live data: Look for anomalies

Possible solution: Heat-sensitive CKP sensor
- Works fine when cool
- Fails when engine hot after 30 min driving
- Hot day makes it worse (underhood temp higher)
```

```
- May not set code if failure brief
- Replace and monitor
```

Multiple Problems Masking Each Other:

Example Scenario:

```
Customer: "Car runs rough and gets poor fuel economy"

Findings during diagnosis:
1. Spark plugs severely worn (one cylinder misfiring)
2. Air filter completely clogged (restricting airflow)
3. Oxygen sensor failed (can't adjust fuel mixture)
4. Dirty throttle body (unstable idle)

Question: Which one is causing the problem?
Answer: ALL of them, working together

Results of fixing only ONE:
- Fix plugs only → still rough (other issues remain)
- Fix air filter only → better but not solved
- Customer frustrated: "You didn't fix it!"

Correct approach:
- Identify ALL problems during initial inspection
- Present complete picture to customer
- Recommend addressing all related issues
- Explain how they interact
- Fix properly, verify results

Result: Car runs properly, customer satisfied, no comeback
```

The "Comeback" Problem:

Comeback = Customer returns with same complaint after repair

Common causes:

1. **Misdiagnosis:** Fixed wrong thing
2. **Incomplete repair:** Fixed symptom, not cause
3. **Missed related problem:** Other issue causing similar symptom
4. **Failed part:** New part defective (rare but happens)
5. **Related damage:** Original problem damaged other parts

How to minimize comebacks:

- Thorough initial diagnosis
- Test after repair
- Check related components
- Use quality parts
- Verify root cause, not symptom
- Document findings clearly

Example:

```
Original complaint: "Battery keeps dying"

Improper diagnosis: Replace battery
Result: Comeback in 2 weeks (battery dead again)

Proper diagnosis sequence:
1. Test battery: Weak, won't hold charge
2. Test charging system: Alternator output low (only 12.8V)
3. Test alternator: Diode failure, not charging properly
4. Check for parasitic drain: Minor dome light draw, acceptable
5. Root cause: Failed alternator not charging battery
6. Repair: Replace alternator AND battery (battery damaged by
undercharging)
7. Verify: Charging voltage 14.2V, battery holds charge
8. Result: No comeback

The key: Battery was symptom, alternator was cause
```

Communication During Complex Diagnosis:

Managing customer expectations:

```
"Mr. Sharma, I've started diagnosing the rough running issue. I've found
three problems so far that are all contributing: severely worn spark
plugs,
a completely clogged air filter, and a failed oxygen sensor. I'm still
checking a few more things because I want to make sure we identify
everything at once rather than fixing one thing and having you come back.

I'll call you in about an hour with a complete report and cost estimate.
The goal is to fix it properly the first time. Does that sound okay?"
```

This approach:

- Sets expectation that diagnosis takes time
- Shows thoroughness (positive)
- Prevents "why didn't you find this the first time?"
- Builds confidence

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Complex Diagnostic Scenario #1

Scenario: Poor Performance with Multiple Symptoms

Vehicle Setup:

- Instructor prepares vehicle with multiple issues:
  - Dirty air filter
  - Disconnected vacuum hose (causing lean condition)
  - One fouled spark plug
  - Low tire pressure (affects "performance" feel)

Customer Complaint (Provided to Students):

```
"My car feels sluggish, especially when accelerating. The check engine
light came on yesterday. It also seems to be using more fuel than normal.
Sometimes it idles roughly when I'm stopped at a light."
```

Student Task: Complete Diagnosis (60 min)

Step 1: Verify Complaint (10 min)

- Read service order
- Note all symptoms customer described
- Create symptom list:
  - Sluggish acceleration
  - Check engine light on
  - Poor fuel economy
  - Rough idle (intermittent)

Step 2: Scan for Codes (10 min)

- Connect OBD-II scanner
- Read codes:

    - P0171 - System Too Lean (Bank 1)
    - P0301 - Cylinder 1 Misfire
- View freeze frame data
- Record live data (RPM, fuel trim, O2 sensors)

Step 3: Systematic Inspection (30 min)

Visual inspection:

- Check air filter:
    - Finding: Severely clogged
    - Impact: Restricts airflow, contributes to poor performance
    - Document with photo
- Inspect engine bay:
    - Finding: Vacuum hose disconnected from intake manifold
    - Impact: Unmetered air entering (causes P0171 lean code)
    - Document location and condition
- Remove and inspect spark plugs:
    - Finding: Cylinder 1 plug heavily fouled (carbon buildup)
    - Impact: Causes misfire (P0301 code), rough idle
    - Other plugs show normal wear
    - Document with photo
- Check tire pressures:
    - Finding: All tires 20 PSI (spec is 32 PSI)
    - Impact: Contributes to sluggish feel, poor economy
    - Not related to codes, but affects performance

Step 4: Create Findings Report (10 min)

Students document:

```
DIAGNOSTIC FINDINGS:

1. CHECK ENGINE LIGHT CODES:
   - P0171: System Too Lean - CAUSE: Disconnected vacuum hose
   - P0301: Cylinder 1 Misfire - CAUSE: Fouled spark plug

2. RELATED PERFORMANCE ISSUES:
   - Clogged air filter - restricts airflow, reduces power
   - Low tire pressure (20 PSI vs 32 spec) - affects handling and economy

3. ROOT CAUSES IDENTIFIED:
   - Vacuum hose disconnected (lean condition, rough idle)
```

```
   - Cylinder 1 spark plug fouled (misfire, rough idle, power loss)
   - Air filter clogged (reduced power, poor economy)
   - Tire pressure low (sluggish feel, poor economy)

4. INTERCONNECTIONS:
   All four issues contribute to customer's complaints. The vacuum
   leak and fouled plug explain the codes and rough idle. The air
   filter and tire pressure explain the performance and economy issues.
```

Instructor Evaluation:

- ✅ Did students identify ALL issues?
- ✅ Did they understand how issues interact?
- ✅ Did they connect symptoms to causes?
- ✅ Did they document findings properly?
- ✅ Did they use systematic approach?

Pass Criteria:

- ✅ Scans and records codes correctly
- ✅ Identifies all four issues
- ✅ Explains how each contributes to symptoms
- ✅ Documents with photos
- ✅ Creates prioritized repair recommendation

Hour 2 (2:00 PM - 3:00 PM): Complex Diagnostic Scenario #2

Scenario: Intermittent Overheating

Vehicle Setup:

- Instructor creates scenario:
  - Coolant level low (not obvious without careful check)
  - Thermostat that sticks intermittently (simulated)
  - Cooling fan relay intermittent connection

Customer Complaint:

```
"My temperature gauge goes into the red sometimes, but not always. It
seems to happen mostly in traffic, and it cools down when I'm on the
highway. I topped up the coolant last week but it seems low again."
```

Student Task: Diagnose Intermittent Overheating (60 min)

Step 1: Verify and Gather Information (10 min)

- Review complaint
- Key observations:
  - Intermittent (not constant)
  - Happens in traffic (low speed/idle)
  - Cools on highway (high speed/airflow)
  - Coolant consumption (indicates leak)
- These clues point to specific systems

Step 2: Systematic Testing (40 min)

Test 1: Coolant Level Check (COLD engine)

- Remove radiator cap (when completely cold)
- Finding: Coolant level 2 inches below full
- Significance: Indicates leak somewhere
- Check overflow tank: Also low
- Document level

Test 2: Pressure Test Cooling System

- Connect pressure tester
- Pump to 1.1 bar (system rating)
- Observe gauge for 5 minutes
- Finding: Pressure drops from 1.1 bar to 0.7 bar
- Conclusion: Leak present
- Inspect for leaks:
  - Check hoses, radiator, water pump
  - Finding: Small leak at water pump weep hole
  - Document with photo

Test 3: Cooling Fan Test

- Start engine, warm to operating temp
- Observe fan operation
- Finding: Fan runs intermittently (starts, stops, starts)
- Check relay: Connection loose
- Wiggle relay: Fan starts/stops
- Conclusion: Poor relay connection causes intermittent fan operation
- This explains: Overheats in traffic (no fan), cools on highway (airflow)

Test 4: Thermostat Function

- Feel upper radiator hose while engine warms
- Should stay cool until thermostat opens (~85°C)
- Then should get hot quickly
- Finding: Hose gets warm gradually (not proper operation)
- Indicates: Thermostat may be stuck partially open or failing
- Note: Partially open = slow warm-up, but shouldn't cause overheating

Test 5: Pressure Test Radiator Cap

- Install cap on tester
- Pump to pressure
- Should release at 1.1 bar
- Finding: Releases properly
- Conclusion: Cap okay

Step 3: Analysis and Recommendation (10 min)

Students create report:

```
INTERMITTENT OVERHEATING - ROOT CAUSES IDENTIFIED:

PRIMARY CAUSES:
1. COOLING FAN - Intermittent Operation
   Cause: Loose relay connection
   Effect: Fan doesn't run consistently in traffic
   When: Overheats at idle/low speed (no airflow, no fan)
   Fix: Clean and secure relay connection OR replace relay
   Cost: ₹500 (relay replacement)

2. COOLANT LEAK - Water Pump
   Cause: Water pump seal leaking at weep hole
   Effect: Coolant level drops, system can't cool effectively
   Customer topped up, but leak continues
   Fix: Replace water pump
   Cost: ₹2,500 parts + ₹1,200 labor = ₹3,700

SECONDARY ISSUE:
3. THERMOSTAT - Possibly Sticking
   Not causing overheating, but not opening properly
   Recommend replacement during water pump service
   (Same labor, minor additional cost)
   Cost: +₹800 parts
```

```
EXPLANATION OF SYMPTOMS:
- Overheats in traffic: Fan not running + low coolant
- Cools on highway: Ram air provides cooling despite fan issue
- Intermittent: Fan works sometimes, doesn't work other times

RECOMMENDATIONS:
🔴 URGENT: Water pump replacement + thermostat + cooling fan relay
Total: ₹5,000 parts + ₹1,500 labor = ₹6,500

BENEFIT: Solves all three issues in one service, prevents engine damage
```

Instructor Discussion:

- Multiple problems contributing to one symptom (overheating)
- Each problem explains part of the behavior
- Customer complaint gives clues (traffic vs highway)
- Systematic testing reveals all issues
- Complete repair prevents comebacks

Pass Criteria:

- ✅ Identifies all three issues
- ✅ Explains how each contributes to symptoms
- ✅ Performs pressure test correctly
- ✅ Tests cooling fan operation
- ✅ Connects findings to customer complaint
- ✅ Recommends complete repair (not piecemeal)

Hour 3 (3:00 PM - 4:00 PM): Complete Pre-Purchase Inspection

Final Integration Exercise: Comprehensive Vehicle Inspection

Scenario:

- Customer considering purchasing used vehicle
- Wants complete inspection before buying
- Student must deliver professional assessment

Vehicle: 2016 Hyundai i20, 82,000 km, ₹4.5 lakh asking price

Student Task: 90-Minute Complete Inspection + Report (60 min)

Inspection checklist (comprehensive):

Exterior (10 min):

- [ ] Body panels (dents, scratches, misalignment)
- [ ] Paint (original, repainted areas, fading)
- [ ] Glass (chips, cracks)
- [ ] Lights (all functional)
- [ ] Tires (tread depth, wear pattern, age)
- [ ] Wheels (damage, curb rash)
- [ ] Under vehicle (rust, damage, fluid leaks)

Under Hood (15 min):

- [ ] Engine oil level and condition
- [ ] Coolant level and condition
- [ ] Brake fluid level and condition
- [ ] Battery condition and voltage
- [ ] Belts (condition, tension)
- [ ] Hoses (cracks, leaks)
- [ ] Air filter
- [ ] Evidence of leaks
- [ ] Engine mounts
- [ ] Wiring condition

Interior (10 min):

- [ ] Seat belts (all positions)
- [ ] Seats (wear, damage, adjustment)
- [ ] Dashboard (cracks, warnings lights)
- [ ] Controls (HVAC, windows, locks, mirrors)
- [ ] Odometer reading (verify with service records)
- [ ] Cleanliness and odor
- [ ] Trunk condition and spare tire

Mechanical (15 min):

- [ ] Suspension (bounce test, leaking shocks)
- [ ] Steering (play, operation)
- [ ] Brakes (pad thickness, rotor condition)
- [ ] Transmission operation (if test drive)
- [ ] Clutch operation (if manual)
- [ ] Exhaust system (leaks, rust, damage)

- [ ] CV boots (tears, leaks)

Electrical/Diagnostic (10 min):

- [ ] Battery voltage (key off, running)
- [ ] Charging system (voltage test)
- [ ] Scan for codes (any stored faults)
- [ ] All lights functional
- [ ] Horn, wipers, washers

Test Drive (if permitted - 10 min):

- [ ] Engine starts easily
- [ ] Acceleration smooth
- [ ] Transmission shifts properly
- [ ] Brakes straight and effective
- [ ] Steering responsive, no pulling
- [ ] No unusual noises or vibrations
- [ ] Warning lights off

Documentation (20 min):

Students create professional pre-purchase inspection report:

```
PRE-PURCHASE INSPECTION REPORT
Vehicle: 2016 Hyundai i20 1.2L Petrol
Odometer: 82,247 km
Asking Price: ₹4,50,000
Inspection Date: [Date]
Inspector: [Student Name]

OVERALL ASSESSMENT: FAIR - Some items need attention before purchase

EXTERIOR CONDITION:
✅ Body: Straight, no accident damage visible
⚠️ Paint: Minor scratches and door dings (normal wear)
⚠️ Front bumper: Small crack on passenger side (₹2,500 repair)
✅ Glass: All intact, no chips
⚠️ Tires:
   - Tread: 4mm all around (adequate but plan replacement soon)
   - Age: 2019 manufactured (5 years old - should replace)
   - Estimated cost: ₹16,000 for set of 4

MECHANICAL CONDITION:
```

```
✅ Engine: Runs smoothly, no unusual noises
✅ Transmission: Shifts properly through all gears
⚠️ Brakes:
   - Front pads: 4mm (adequate for 15,000 km)
   - Rear pads: 6mm (good)
   - Rotors: Light grooves, within spec
⚠️ Suspension:
   - Front shocks: Slight seepage (not failed, but aging)
   - Bushings: Some deterioration visible
   - Recommend monitoring, may need in 20,000 km

FLUIDS & MAINTENANCE:
✅ Engine oil: Clean, recent change evident
⚠️ Coolant: Original (never changed in 82,000 km)
   - Recommend flush and replacement (₹2,000)
⚠️ Brake fluid: Dark (old)
   - Recommend replacement (₹1,200)
✅ Transmission: Fluid appears good

ELECTRICAL:
✅ Battery: 12.6V (good charge), 2 years old
✅ Charging: 14.1V (alternator working properly)
✅ All lights functional
⚠️ Check engine light: One stored code (P0420 - Catalyst efficiency)
   - Not currently illuminated
   - May indicate future exhaust expense (₹15,000-25,000)
   - Recommend OBD-II test drive monitoring

INTERIOR:
✅ Seats: Good condition, no tears
✅ Dashboard: No cracks
⚠️ A/C: Blows cool but not very cold (may need recharge - ₹2,500)
✅ All controls functional

ITEMS REQUIRING IMMEDIATE ATTENTION:
1. Coolant system service: ₹2,000
2. Brake fluid replacement: ₹1,200
3. Front bumper crack repair: ₹2,500
Total: ₹5,700

ITEMS TO BUDGET FOR (6-12 months):
1. Tires (age/tread): ₹16,000
2. A/C service: ₹2,500
```

```
3. Possible catalyst (if code returns): ₹15,000-25,000
4. Front shocks (if leaking worsens): ₹8,000

NEGOTIATION LEVERAGE:
- Immediate repairs needed: ₹5,700
- Near-term expenses: ₹26,500-36,500
- Total: ₹32,200-42,200

RECOMMENDATION:
The vehicle is in fair mechanical condition with no major issues found.
However, deferred maintenance and aging components will require
investment.
I recommend offering ₹4,20,000 (₹30,000 below asking) to account for
immediate and near-term maintenance needs.

If purchased:
- Perform coolant and brake fluid service immediately
- Budget for tires within 6 months
- Monitor catalyst code (may need future repair)

The vehicle appears to have been maintained adequately but not
excellently.
With proper care going forward, it should provide reliable transportation.

Inspector Signature: _________________ Date: _______

Photos attached: [15 photos documenting findings]
```

Instructor Reviews Reports (30 min):

- Each student presents findings
- Discuss assessment
- Evaluate professionalism
- Provide feedback

Pass Criteria:

- ✅ Completes systematic inspection
- ✅ Documents all findings with photos
- ✅ Prioritizes issues correctly
- ✅ Provides realistic cost estimates
- ✅ Makes clear recommendation
- ✅ Professional report format

- ✅ Explains technical findings clearly
- ✅ Considers buyer's perspective

DAY 29 ASSESSMENT:

Practical Skills Checklist:

- [ ] Applies systematic diagnostic methodology
- [ ] Scans and interprets fault codes
- [ ] Performs multiple system inspection
- [ ] Identifies all problems (doesn't stop at first finding)
- [ ] Understands system interactions
- [ ] Documents findings with photos
- [ ] Creates prioritized repair recommendations
- [ ] Performs complete pre-purchase inspection
- [ ] Creates professional inspection report
- [ ] Makes evidence-based recommendations

Knowledge Check (Verbal):

- What are the 8 steps of systematic diagnosis?
- Why is it important to identify all problems, not just the first one?
- Give an example of how two systems can interact to cause one symptom
- What should be included in a pre-purchase inspection report?
- How do you prioritize multiple findings?

DAY 30: FINAL ASSESSMENT + CAREER GUIDANCE

Day Objective:

- Evaluate student competency through written and practical assessments
- Provide career path guidance
- Overview specialization tracks
- Course completion and certification

MORNING SESSION: FINAL ASSESSMENTS (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Final Written Assessment

Written Exam Structure:

- **Total questions:** 100
- **Time allowed:** 60 minutes
- **Pass mark:** 70% (70/100 correct)
- **Safety questions:** Must score 100% on safety (10 questions)
- **Format:** Multiple choice, matching, short answer

Content Distribution:

Week 1: Vehicle & Engine Fundamentals (20 questions)

- Workshop safety (5 questions) - CRITICAL
- Vehicle structure and systems
- Engine operation (4-stroke cycle)
- Intake and exhaust systems
- Basic tool identification
- Measurement units

Week 2: Engine Support & Power Delivery (20 questions)

- Emission control systems
- Cooling system operation
- Lubrication system
- Clutch system
- Transmission types and operation
- Preventive maintenance schedules

Week 3: Chassis, Control & Safety (25 questions)

- Suspension components and types
- Steering system operation
- Brake system operation
- ABS basics
- Tire specifications and inspection
- HVAC systems
- Safety systems (airbags, seat belts)

Week 4: Electrical Systems (20 questions)

- Basic electrical theory
- Battery specifications and testing
- Starting and charging systems
- Wiring identification

- Sensor and actuator identification
- OBD-II system and scanner operation

Week 5: Professional Practices (15 questions)

- Service schedules interpretation
- 5S methodology
- Torque specifications importance
- Tool management
- Customer communication
- Documentation requirements

Sample Questions:

Safety (Must get 100% - 10 questions):

1. Before lifting a vehicle, you must: (a) Check lift points match vehicle, (b) Just lift it quickly, (c) Don't need to check anything
2. When working near airbags, you must: (a) Disconnect battery and wait 10 minutes, (b) Work immediately, (c) Airbags are safe to work around
3. Used engine oil should be: (a) Poured down drain, (b) Recycled properly, (c) Thrown in trash
4. When jump-starting a vehicle, connect the negative cable to: (a) Dead battery negative terminal, (b) Engine block of dead vehicle, (c) Doesn't matter
5. Yellow connectors in a vehicle indicate: (a) Airbag circuits, (b) Regular circuits, (c) High voltage
6. A torque wrench should be stored: (a) At highest setting, (b) At lowest setting, (c) Any setting is fine
7. If brake fluid gets on paint: (a) Leave it, (b) Wipe immediately, it damages paint, (c) Wash later
8. When checking tire pressure: (a) Tires should be hot, (b) Tires should be cold, (c) Doesn't matter
9. A cracked wrench should be: (a) Used carefully, (b) Removed from service immediately, (c) Welded and reused
10. Before removing a radiator cap: (a) Engine must be completely cold, (b) Can remove when warm, (c) Temperature doesn't matter

Technical Questions (Sample):

Engine Fundamentals: 11. The four strokes of a 4-stroke engine are: (a) Intake, Compression, Power, Exhaust, (b) Intake, Power, Compression, Exhaust, (c) Start, Run, Stop, Rest

12. Which component controls airflow into the engine? (a) Fuel injector, (b) Throttle body, (c) Oxygen sensor

Cooling System: 13. Why is the cooling system pressurized? (a) To make it work better, (b) To raise the boiling point of coolant, (c) To reduce coolant usage

14. Normal engine operating temperature is: (a) 50-60°C, (b) 85-95°C, (c) 120-130°C

Brakes: 15. Minimum legal brake pad thickness in most areas is: (a) 1.6mm, (b) 5mm, (c) 10mm

16. What does ABS prevent? (a) Brake fade, (b) Wheel lockup, (c) Brake wear

Electrical: 17. A fully charged battery should read: (a) 10.5V, (b) 12.6V, (c) 14.5V

18. Normal charging system voltage is: (a) 12.6V, (b) 13.5-14.5V, (c) 15.5V

Tires: 19. In the tire size 205/55R16, the "55" represents: (a) Width in mm, (b) Aspect ratio (sidewall height %), (c) Rim diameter

20. Legal minimum tire tread depth is: (a) 1.6mm, (b) 3mm, (c) 5mm

Professional Practices: 21. Service recommendations should be prioritized as: (a) Random order, (b) Urgent/Soon/Monitor, (c) Most expensive first

22. The 5S methodology focuses on: (a) Workplace organization and quality, (b) Speed of service, (c) Sales techniques

Students complete exam in 60 minutes Exams collected and graded during Hour 2

Hour 2 (10:00 AM - 11:00 AM): Final Practical Assessment Setup

Practical Assessment Structure:

- **Stations:** 6 practical stations (10 minutes each)
- **Total time:** 60 minutes
- **Pass mark:** 70% overall (100% on safety-critical tasks)
- **Format:** Hands-on demonstration of skills

Station 1: Vehicle Lifting & Wheel Service (10 min) Skills tested:

- Identifies correct lift points
- Positions lift pads correctly
- Lifts vehicle safely
- Installs wheel using proper sequence (star pattern)

- Torques lug nuts to specification
- Returns torque wrench to lowest setting

Evaluation:

- [ ] Correct lift points identified
- [ ] Lifts safely (no vehicle movement)
- [ ] Star pattern used
- [ ] Torque wrench set correctly
- [ ] Torques all lug nuts
- [ ] Stores torque wrench properly **Safety critical:** Lifting procedure (must be perfect)

Station 2: Fluid Level Checks (10 min) Skills tested:

- Checks engine oil level correctly
- Checks coolant level (engine cold only)
- Checks brake fluid level
- Checks windshield washer fluid
- Assesses fluid conditions (oil color, coolant clarity)

Evaluation:

- [ ] Oil dipstick used correctly (wipe, reinsert, read)
- [ ] Coolant checked when cold
- [ ] Brake fluid level read correctly
- [ ] Assesses oil condition
- [ ] Notes any low fluids **Safety critical:** Only checks coolant when cold

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Station 3: Battery & Electrical Testing (10 min) Skills tested:

- Tests battery voltage (engine off)
- Tests charging voltage (engine running)
- Cleans battery terminals
- Performs continuity test on fuse
- Identifies blown vs good fuse visually

Evaluation:

- [ ] Multimeter set to DC voltage
- [ ] Probes connected correctly (red to +, black to -)

- [ ] Engine off voltage recorded
- [ ] Engine running voltage recorded
- [ ] Interprets readings correctly
- [ ] Fuses identified correctly **Safety critical:** Correct multimeter connections

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Station 4: Brake Inspection & Measurement (10 min) Skills tested:

- Measures brake pad thickness with caliper
- Inspects brake rotor condition
- Checks brake hose condition
- Records measurements accurately

Evaluation:

- [ ] Uses caliper correctly (zeroed, proper measurement)
- [ ] Measures thinnest point of pad
- [ ] Records measurement in mm
- [ ] Compares to minimum (3mm)
- [ ] Inspects rotor for grooves/cracks
- [ ] Identifies worn vs good brake components **Safety critical:** Accurate measurement (not visual guess)

Station 5: OBD-II Scanner Operation (10 min) Skills tested:

- Locates diagnostic connector
- Connects scanner correctly
- Reads fault codes
- Views freeze frame data
- Clears codes
- Checks monitor status

Evaluation:

- [ ] DLC located quickly
- [ ] Scanner connected properly
- [ ] Communication established
- [ ] Codes read and recorded
- [ ] Freeze frame accessed
- [ ] Codes cleared properly **Safety critical:** None (but must demonstrate competency)

Station 6: Multi-Point Inspection & Documentation (10 min) Skills tested:

- Performs systematic under-hood inspection
- Checks tire tread depth
- Checks tire pressure
- Documents findings on checklist
- Takes photo of one finding

Evaluation:

- [ ] Systematic approach (doesn't skip items)
- [ ] Tread depth measured correctly
- [ ] Tire pressure checked and adjusted
- [ ] Findings documented clearly
- [ ] Photo shows problem clearly
- [ ] Identifies urgent vs monitor items **Safety critical:** None (but demonstrates professional competency)

Practical Assessment Execution:

Student rotation:

- Groups of 6 students
- Each starts at different station
- 10 minutes per station
- Rotate on signal
- Instructor observes and scores

Scoring:

- Each station worth equal points
- Safety-critical tasks must be perfect
- Overall 70% required to pass
- Remediation available for failures

Hour 3 (11:00 AM - 12:00 PM): Assessment Grading & Break

Instructors grade assessments:

- Written exams scored
- Practical station scores tallied
- Individual student performance reviewed
- Identify any students needing remediation

Students take break:

- Relax after assessments
- Informal discussion about course
- Prepare for afternoon session

AFTERNOON SESSION: CAREER GUIDANCE (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Career Paths in Automotive Service

The Automotive Industry Landscape:

Industry Growth:

- India's automotive market expanding rapidly
- Increasing vehicle complexity (electronics, emissions)
- Shortage of skilled technicians
- **High demand for trained professionals**
- **Job security in this field**

Career Paths for Automotive Technicians:

1. Independent Workshop Technician

- Work in local repair shop
- Variety of vehicles and problems
- Direct customer interaction
- Opportunity to build reputation
- **Salary range:** ₹15,000-35,000/month (with experience)
- **Pros:** Diverse work, local relationships, flexibility
- **Cons:** Long hours, variable income, customer complaints

2. Dealership Technician

- Manufacturer-authorized service center
- Specialize in one brand
- Factory training and tools
- Career ladder (Level 1 → Master Technician)

- **Salary range:** ₹18,000-50,000/month (with experience)
- **Pros:** Structured environment, brand training, benefits
- **Cons:** Repetitive work, warranty time limits, corporate policies

3. Fleet Maintenance Technician

- Commercial vehicle fleets (trucks, buses, taxis)
- Preventive maintenance focused
- Industrial/commercial setting
- **Salary range:** ₹20,000-40,000/month
- **Pros:** Predictable schedule, preventive work, less customer interaction
- **Cons:** Heavy vehicle work, industrial environment

4. Specialty Shop Technician

- Performance tuning
- Diesel specialist
- Transmission specialist
- Electrical specialist
- **Salary range:** ₹25,000-60,000/month (highly skilled)
- **Pros:** Expert in specific area, premium pay, interesting work
- **Cons:** Requires extensive additional training, limited job market

5. Mobile Mechanic

- Travel to customer locations
- Convenience service
- Higher rates possible
- Self-employed opportunity
- **Income potential:** ₹30,000-80,000/month (experienced, established)
- **Pros:** Independence, flexibility, premium pricing
- **Cons:** Equipment investment, travel time, weather dependent

6. Service Advisor/Writer

- Transition from technical to service management
- Customer interface role
- Sells service, coordinates with technicians
- Commission-based income potential
- **Salary range:** ₹20,000-50,000/month + commission
- **Pros:** Customer-facing, sales opportunity, less physical
- **Cons:** Requires customer service skills, stress from unhappy customers

7. Shop Foreman/Manager

- Oversee technicians
- Quality control
- Workflow management
- Still perform technical work
- **Salary range:** ₹35,000-70,000/month
- **Pros:** Leadership role, higher pay, varied responsibilities
- **Cons:** Management stress, responsible for others' work

8. Training Instructor

- Teach at technical schools
- Share knowledge with new generation
- Academic environment
- **Salary range:** ₹25,000-50,000/month
- **Pros:** Teaching, stable schedule, satisfaction of mentoring
- **Cons:** Lower pay than senior technician, requires teaching skills

Skills That Increase Earning Potential:

- **Specialization:** Electrical, diagnostics, diesel, performance
- **Certifications:** Manufacturer certifications, ASE (international)
- **Customer service:** Communication increases repeat business
- **Continuous learning:** Stay current with technology
- **Business skills:** For shop owners or service advisors

What Employers Look For:

- **Technical competence** (obviously)
- **Problem-solving ability** (diagnostic thinking)
- **Work ethic** (reliable, punctual, thorough)
- **Communication skills** (explain to customers)
- **Honesty and integrity** (customer trust)
- **Willingness to learn** (technology changes constantly)
- **Professional appearance** (clean uniform, organized)

Hour 2 (2:00 PM - 3:00 PM): Specialization Track Overview

The Three Specialization Tracks:

Based on the Curriculum Separation Document:

TRACK 1: MECHANICAL SPECIALIZATION

Focus: Precision mechanical work, engine building, component overhaul

Who should choose this track:

- Enjoy hands-on mechanical work
- Like precision and detail
- Interested in engine internals
- Patient with complex assemblies
- Good with measurements and specifications

What you'll learn:

- **Precision measurement tools:**
  - Micrometers (0.01mm precision)
  - Dial indicators
  - Bore gauges
  - Telescoping gauges
  - Plastigage for bearing clearances
- **Engine rebuilding:**
  - Complete disassembly/assembly
  - Measuring bore, stroke, compression ratio
  - Selecting pistons, rings, bearings
  - Valve train assembly
  - Timing chain/belt installation
  - Torque sequences and specifications
- **Precision mechanical skills:**
  - Cylinder head service (valve grinding, seat cutting)
  - Crankshaft inspection and installation
  - Camshaft timing
  - Thread repair (helicoil, time-sert)
  - Honing and boring
  - Balancing (basic awareness)
- **Transmission overhaul:**
  - Manual transmission rebuild
  - Automatic transmission basics
  - Clutch replacement and adjustment
  - CV joint/axle service
- **Heavy component work:**

    - Engine removal and installation
    - Transmission removal and installation
    - Subframe work

Career opportunities:

- Engine builder
- Machine shop technician
- Performance shop specialist
- Diesel mechanic
- Heavy equipment technician

Duration: 6 months intensive training after fundamentals

TRACK 2: ELECTRICAL/ELECTRONICS SPECIALIZATION

Focus: Wiring repair, advanced diagnostics, module programming

Who should choose this track:

- Enjoy problem-solving and detective work
- Comfortable with electronics
- Patient with intermittent problems
- Like learning new technology
- Detail-oriented with wiring

What you'll learn:

- **Advanced wiring skills:**
    - Wire splicing and soldering
    - Connector repair and pin replacement
    - Harness fabrication
    - Crimping techniques
    - Heat shrink and protection
- **Advanced electrical testing:**
    - Oscilloscope use
    - Current draw testing
    - Voltage drop testing (advanced)
    - Parasitic drain diagnosis
    - Circuit analysis
- **Electronic systems:**

    - Body control modules
    - Instrument clusters
    - Infotainment systems
    - Comfort and convenience systems
    - Security/immobilizer systems
    - CAN bus communication
- **Module programming:**
    - Key programming
    - Module initialization
    - Software updates
    - Configuration and coding
- **Advanced lighting:**
    - LED and HID systems
    - Adaptive headlights
    - Aftermarket installations
- **Accessory installation:**
    - Remote start systems
    - Alarm systems
    - Audio system upgrades
    - Trailer wiring

Career opportunities:

- Auto electrician specialist
- Module programming technician
- Automotive electronics installer
- High-end audio installer
- Fleet electrical specialist

Duration: 6 months intensive training after fundamentals

TRACK 3: DIAGNOSTICS SPECIALIZATION

Focus: Root cause analysis, advanced scan tool use, complex problem-solving

Who should choose this track:

- Excellent problem-solver
- Enjoy detective work
- Comfortable with computers
- Patient with difficult problems

- Like working through complex issues logically

What you'll learn:

- **Advanced scan tool operation:**
  - Bidirectional controls
  - Data graphing and analysis
  - Comparing parameters to known-good
  - Guided diagnostics
  - Manufacturer-specific software
- **Diagnostic methodology:**
  - Systematic troubleshooting
  - Root cause analysis
  - Creating diagnostic trees
  - Using service information effectively
  - TSB research and application
- **Advanced sensor testing:**
  - Oscilloscope patterns
  - Signal analysis
  - Sensor comparison testing
  - Reference voltage verification
- **Drivability diagnosis:**
  - Fuel trim analysis
  - Misfire diagnosis (pattern analysis)
  - Performance complaints
  - Emission failures
  - Intermittent problems
- **Network diagnostics:**
  - CAN bus testing
  - Module communication issues
  - Network troubleshooting
- **ADAS calibration (awareness):**
  - Camera calibration
  - Radar calibration
  - After windshield replacement
  - After suspension work
- **Hybrid/EV systems (introduction):**
  - High-voltage safety
  - Hybrid system operation
  - Battery testing

- Regenerative braking

Career opportunities:

- Diagnostic technician (highest-paid tech role)
- Master technician at dealership
- Drivability specialist
- Training instructor
- Shop foreman/manager

Duration: 6 months intensive training after fundamentals

Choosing Your Path:

Self-Assessment Questions:

1. **What type of work do you enjoy most?**
   - Hands-on mechanical = Mechanical Track
   - Problem-solving and detective work = Diagnostics Track
   - Wiring and electronics = Electrical Track
2. **What are your strengths?**
   - Precision and patience = Mechanical
   - Logical thinking = Diagnostics
   - Attention to detail with circuits = Electrical
3. **What environment do you prefer?**
   - Heavy work, engines = Mechanical
   - Computer-based, thinking = Diagnostics
   - Mix of both = Electrical
4. **What's your career goal?**
   - Engine builder, performance = Mechanical
   - Highest-paid technician = Diagnostics
   - Specialty installer, programming = Electrical

You can eventually learn multiple tracks:

- Most technicians specialize first, then cross-train
- "T-shaped" technician: Deep in one area, broad in others
- Increases versatility and value

Hour 3 (3:00 PM - 4:00 PM): Results, Certification & Course Completion

Assessment Results Distribution (30 min)

Individual meetings with each student:

- Written exam score
- Practical assessment score
- Overall grade
- Feedback on performance

Result categories:

PASS (70%+ overall, 100% safety):

- Congratulations on completion
- Certificate of Completion awarded
- Eligible for specialization track
- Recommendation for employment

CONDITIONAL PASS (70%+ but failed safety items):

- Must remediate safety topics
- Retest on safety questions
- Certificate issued after passing safety

FAIL (Below 70%):

- Review areas of weakness
- Additional study recommended
- Retest scheduled (within 1 week)
- Support provided for improvement